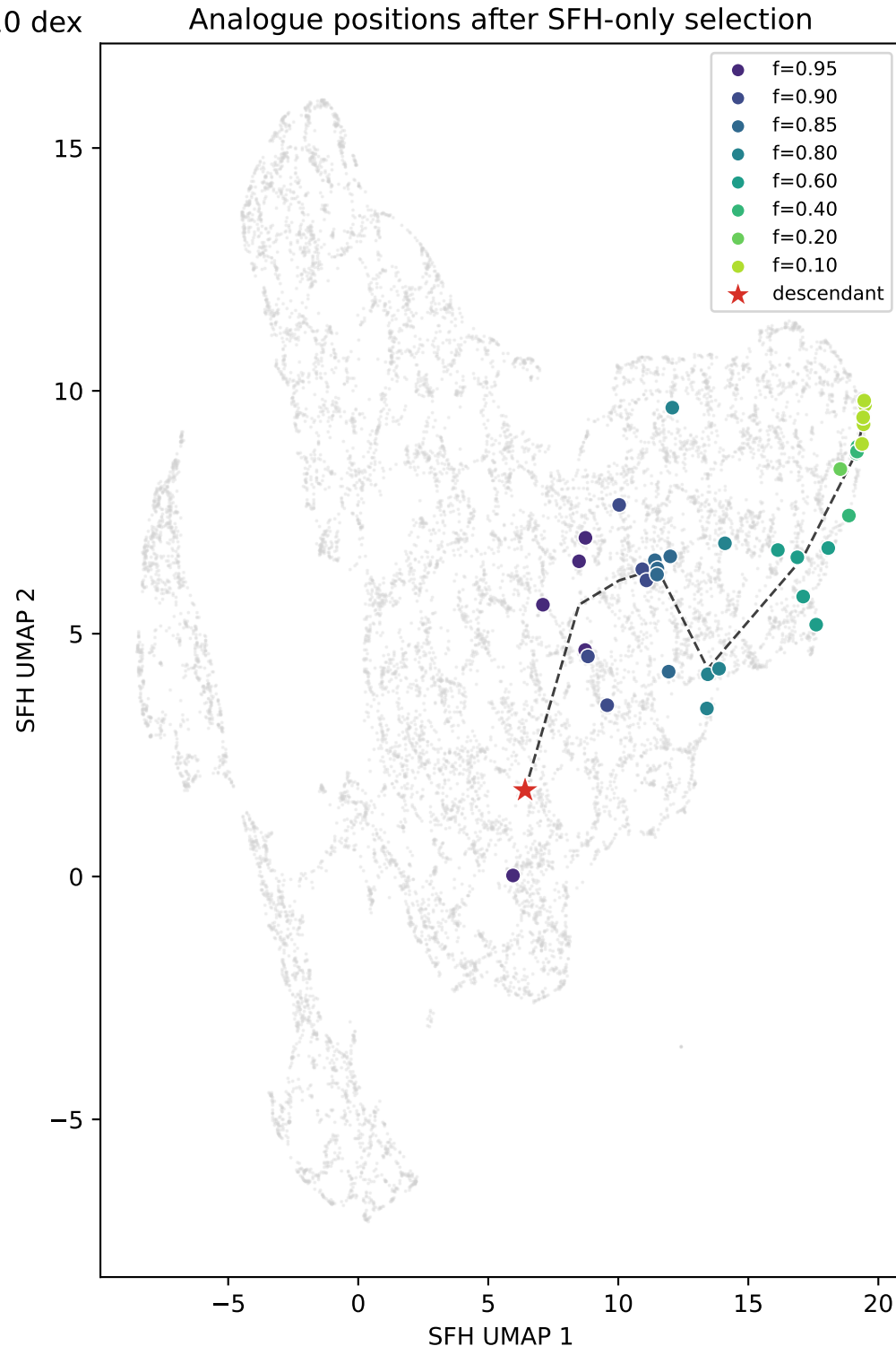
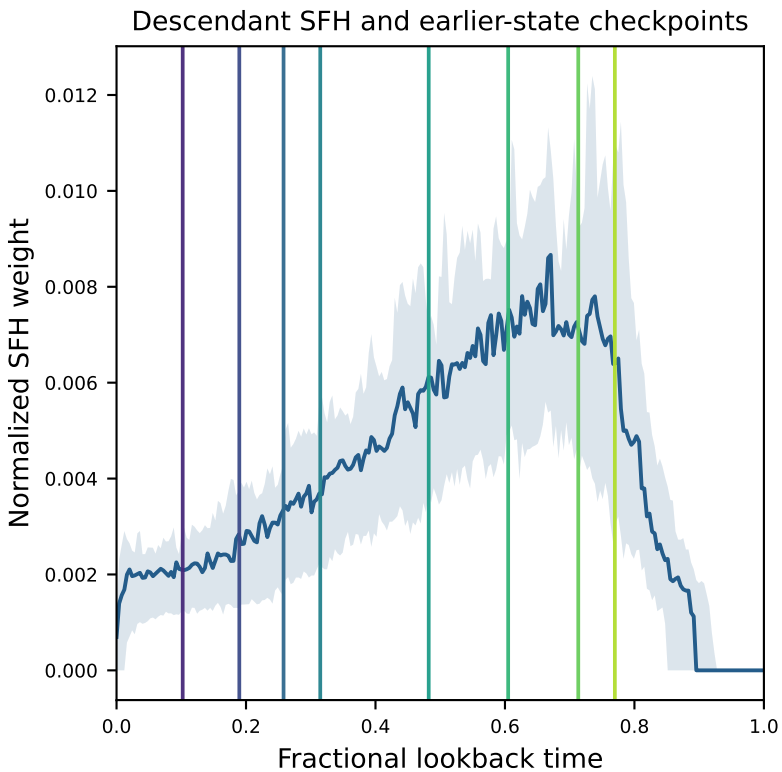
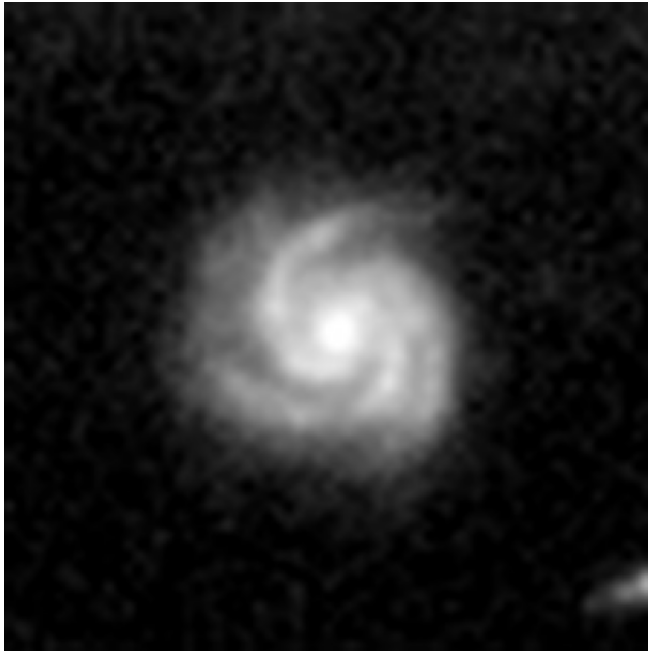
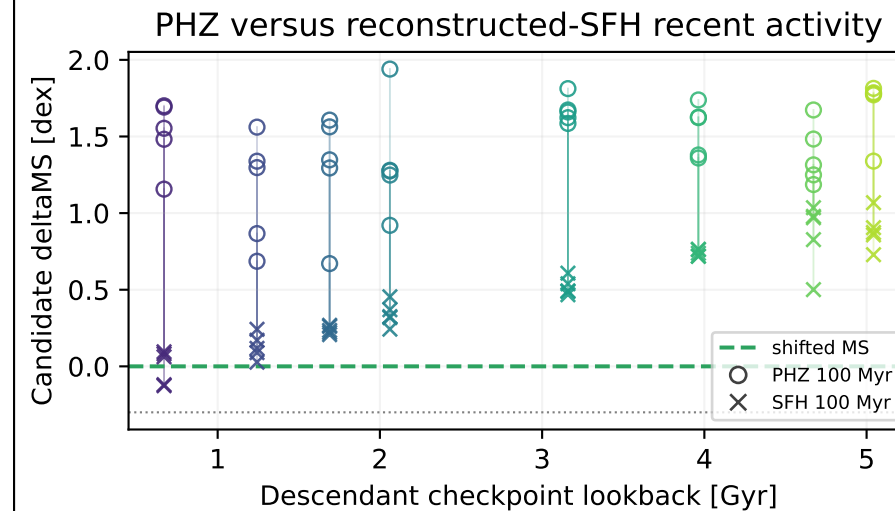


Pilot SFH-selected progenitor-analogue sequence

Descendant 2710874228641049720
 $\log M^*=10.34$, $z=0.818$, $\log \text{sSFR}=-8.23$
 PHZ $\Delta\text{MS}=+1.90$; SFH-100Myr $\Delta\text{MS}=-0.10$ dex



f	L	logM	T	N	zmed
0.95	0.67	10.32	2.00	1,989	0.54
0.90	1.24	10.30	2.00	1,996	0.54
0.85	1.69	10.27	2.00	2,016	0.54
0.80	2.06	10.25	2.00	2,019	0.53
0.60	3.16	10.12	2.00	2,024	0.52
0.40	3.96	9.95	2.00	2,050	0.45
0.20	4.67	9.64	1.88	1,737	0.37
0.10	5.04	9.34	1.51	1,529	0.33



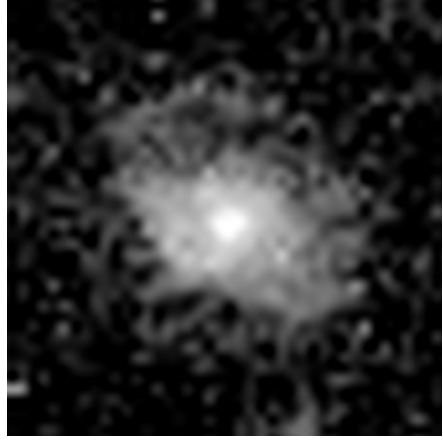
Selection variables

- predicted stellar mass, within ± 0.15 dex
- cumulative SFH before each descendant state
- 2 Gyr physical-time comparison when available

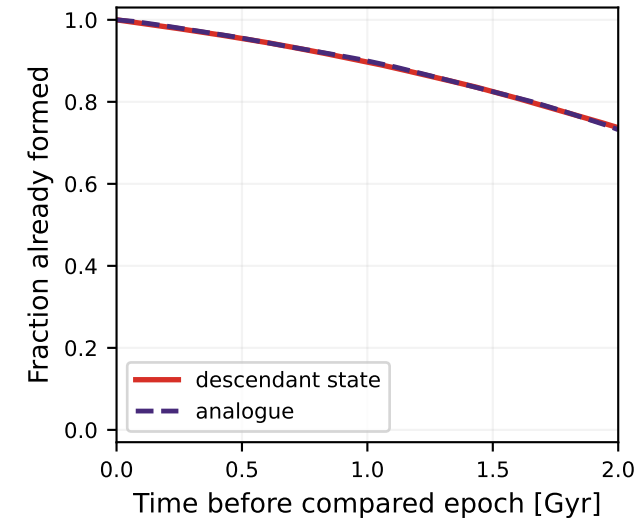
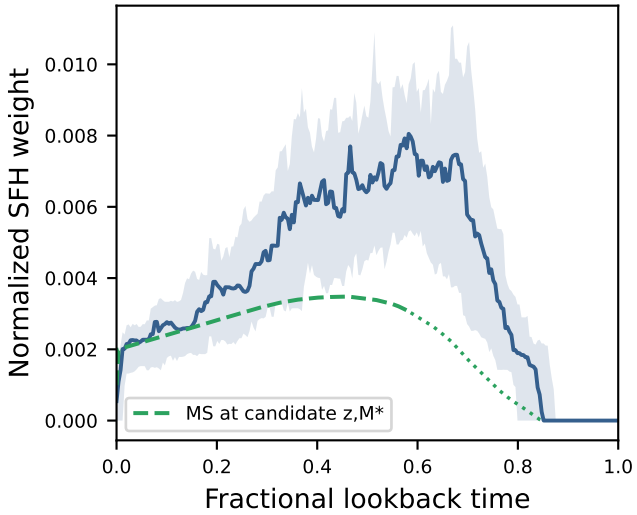
Candidate redshift, morphology, ΔMS , image embedding, and UMAP position are outcomes, not selection variables. The dashed path joins ensemble medians, not one galaxy's orbit.

Earlier state $f=0.95$: lookback=0.67 Gyr, predicted $\log M^*=10.32$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

rank 1: 2758528497640400544
 $\log M^*=10.45 (+0.13)$ $z=0.934$
 PHZ $\Delta MS=+1.48$; SFH-100Myr $\Delta MS=-0.12$
 $D=0.001$



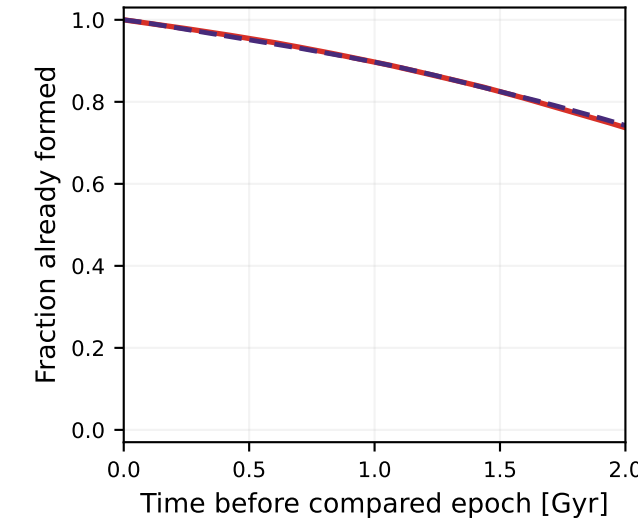
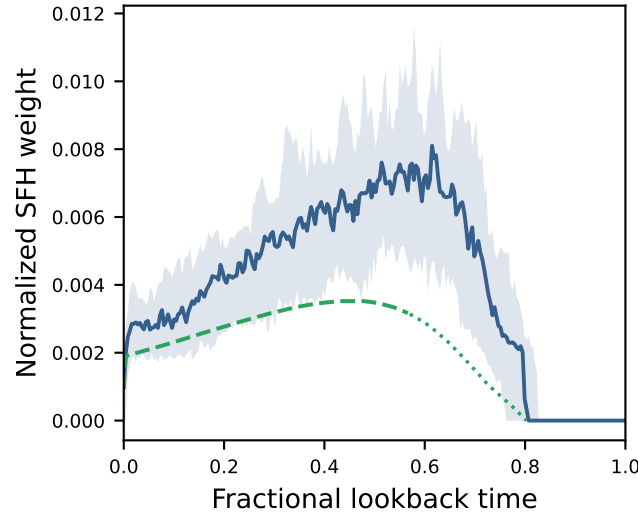
Complete candidate SFH



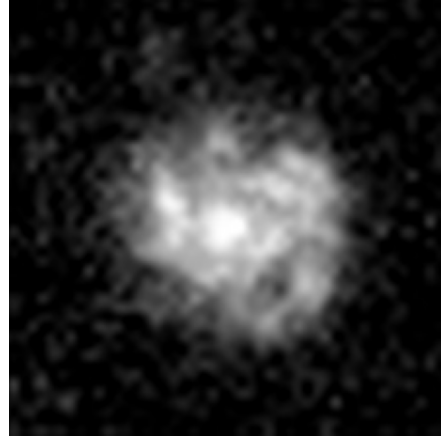
rank 2: 2758797559676788642
 $\log M^*=10.24 (-0.09)$ $z=0.724$
 PHZ $\Delta MS=+1.16$; SFH-100Myr $\Delta MS=+0.08$
 $D=0.002$



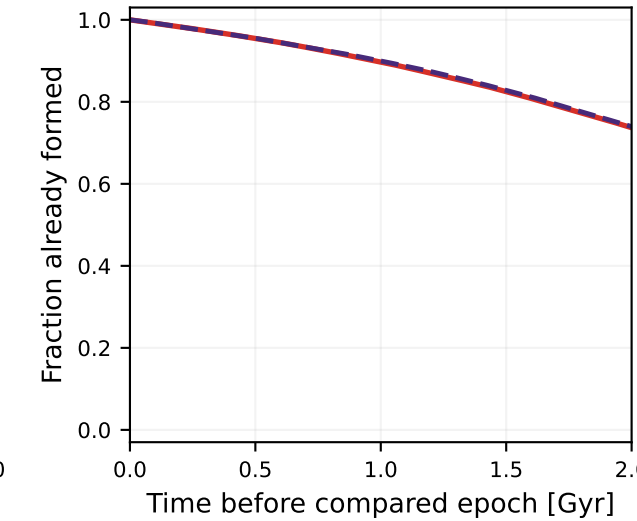
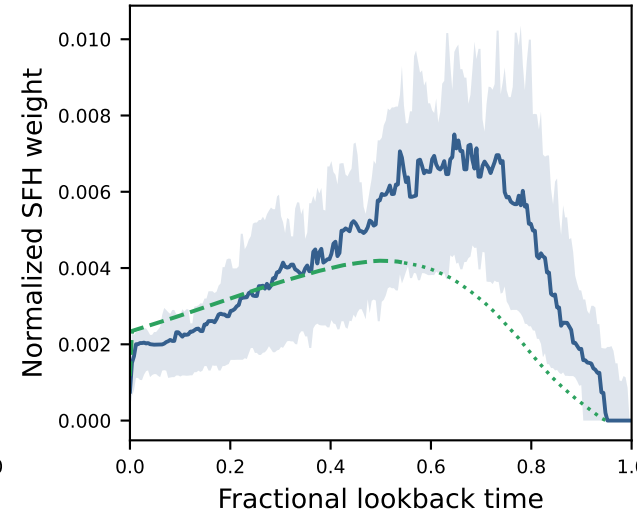
Complete candidate SFH



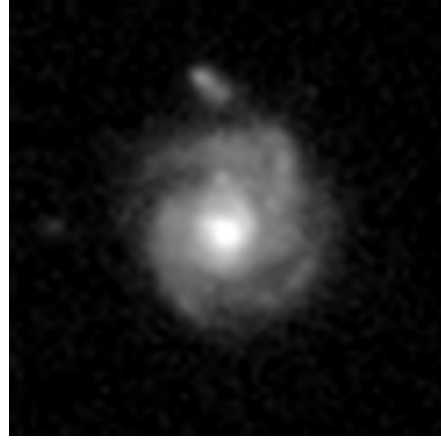
rank 3: 2750960247643309266
 $\log M^*=10.37 (+0.05)$ $z=1.068$
 PHZ $\Delta MS=+1.69$; SFH-100Myr $\Delta MS=-0.13$
 $D=0.002$



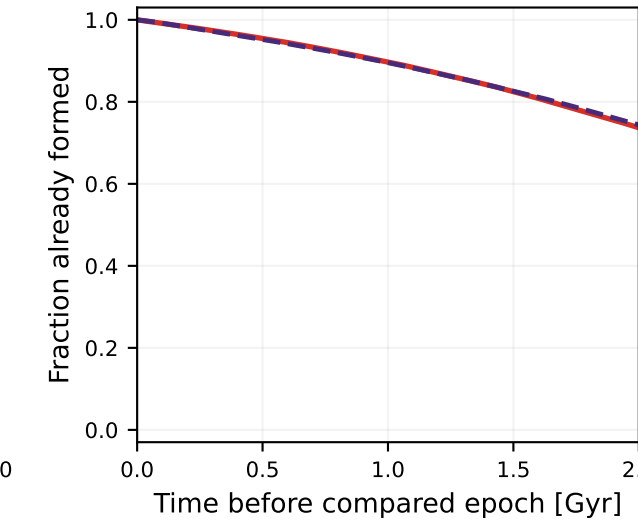
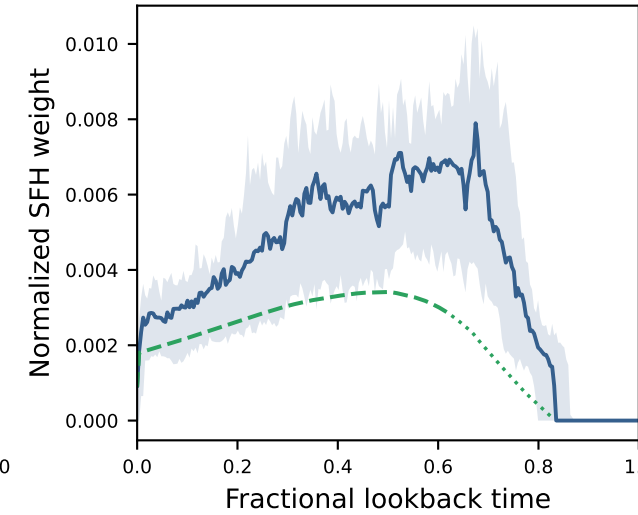
Complete candidate SFH



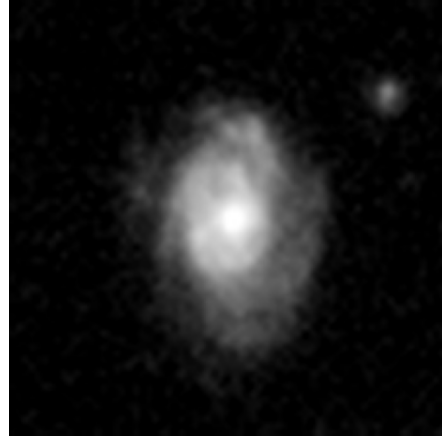
rank 4: 2714848959639851096
 $\log M^*=10.34 (+0.02)$ $z=0.746$
 PHZ $\Delta MS=+1.55$; SFH-100Myr $\Delta MS=+0.09$
 $D=0.002$



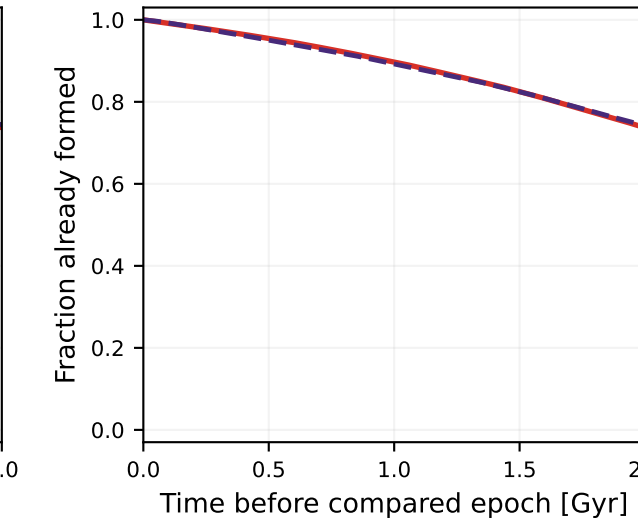
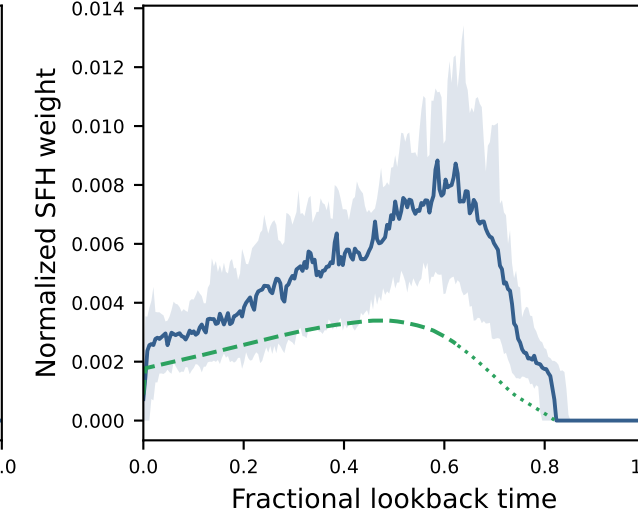
Complete candidate SFH



rank 5: 2678340320651339065
 $\log M^*=10.41 (+0.09)$ $z=0.786$
 PHZ $\Delta MS=+1.70$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.003$

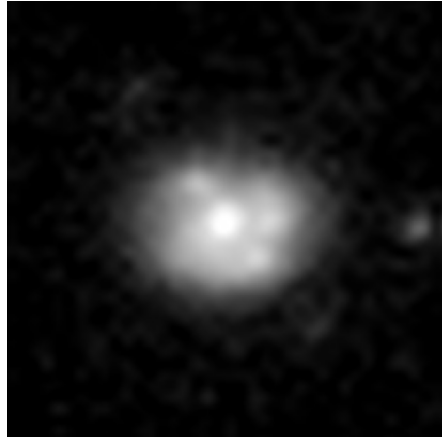


Complete candidate SFH

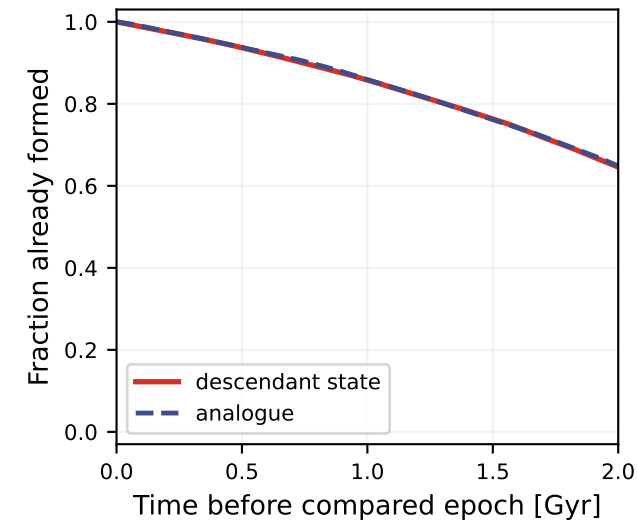
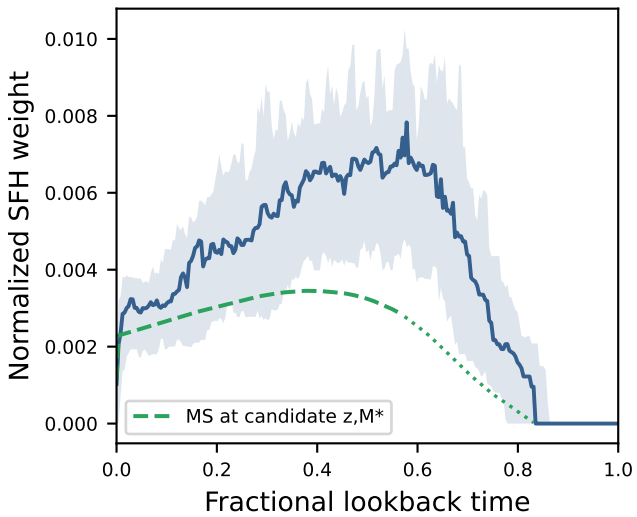


Earlier state $f=0.90$: lookback=1.24 Gyr, predicted $\log M^*=10.30$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

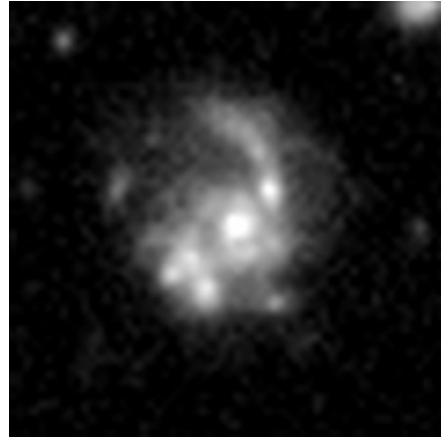
rank 1: 2642114313645187717
 $\log M^*=10.30$ (+0.00) $z=0.968$
 PHZ $\Delta MS=+1.56$; SFH-100Myr $\Delta MS=+0.03$
 $D=0.001$



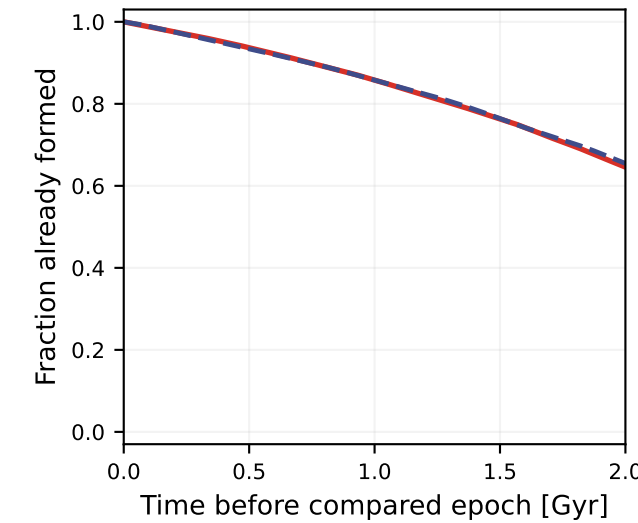
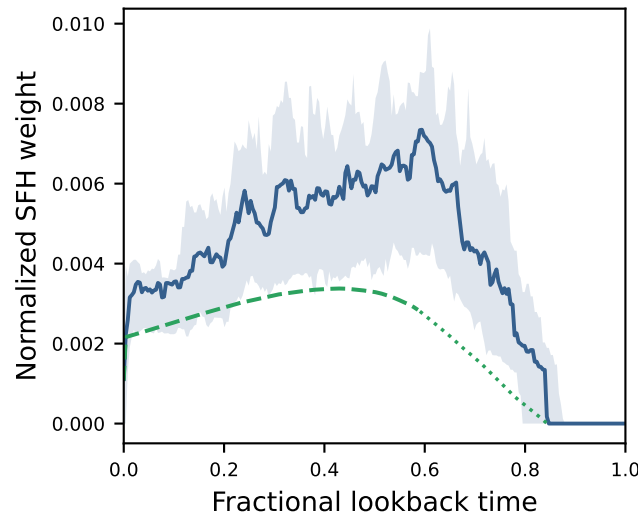
Complete candidate SFH



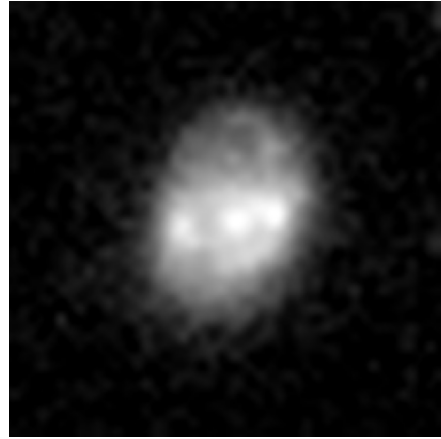
rank 2: 2736389342656361928
 $\log M^*=10.32$ (+0.02) $z=0.912$
 PHZ $\Delta MS=+1.34$; SFH-100Myr $\Delta MS=+0.09$
 $D=0.002$



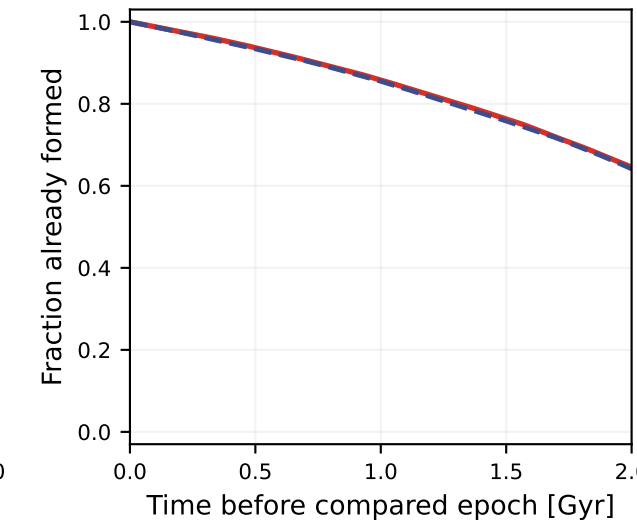
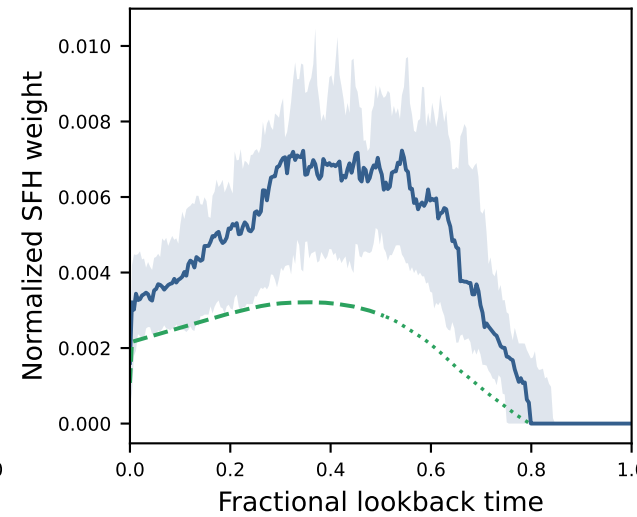
Complete candidate SFH



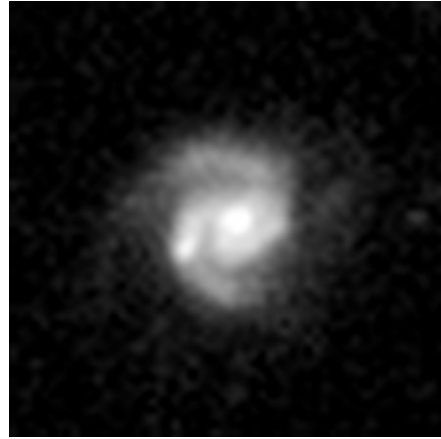
rank 3: 2629667783659961275
 $\log M^*=10.22$ (-0.08) $z=0.834$
 PHZ $\Delta MS=+0.69$; SFH-100Myr $\Delta MS=+0.17$
 $D=0.003$



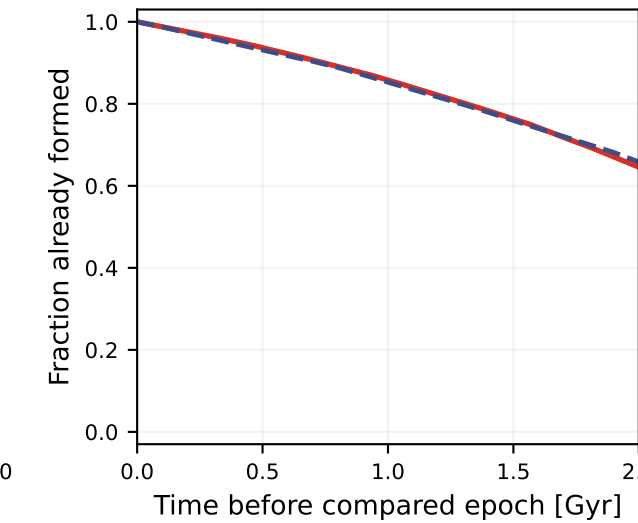
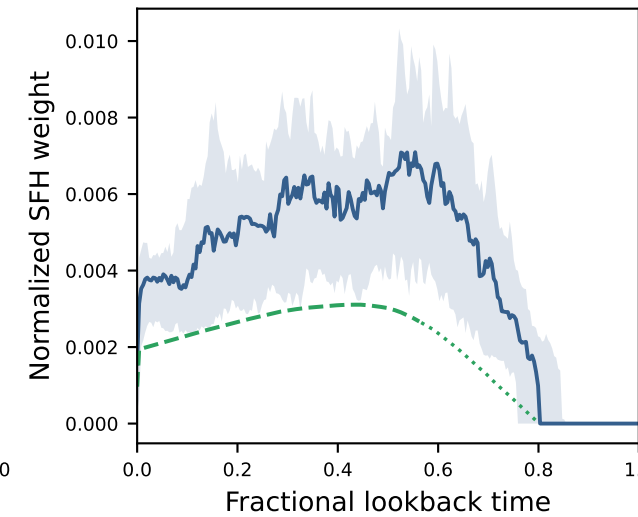
Complete candidate SFH



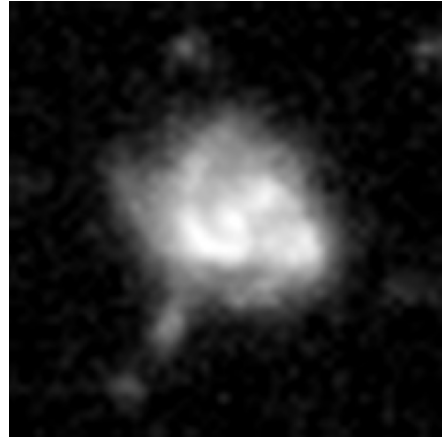
rank 4: 2730582075671504940
 $\log M^*=10.30$ (+0.00) $z=0.790$
 PHZ $\Delta MS=+0.87$; SFH-100Myr $\Delta MS=+0.24$
 $D=0.005$



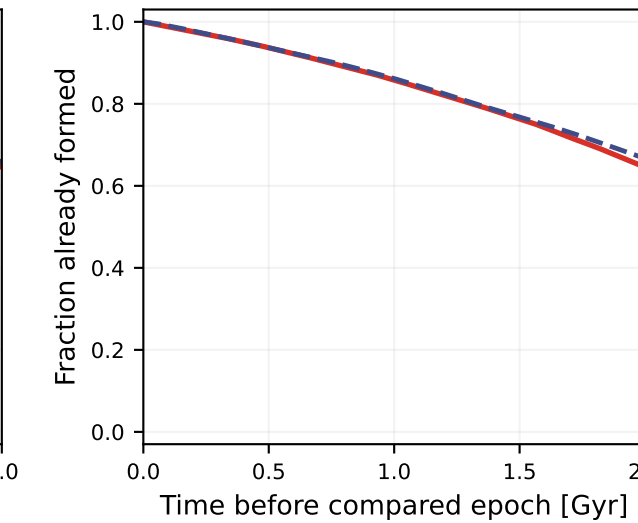
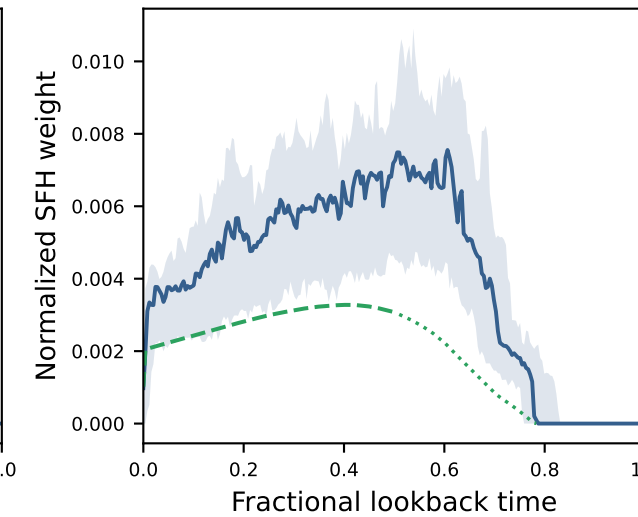
Complete candidate SFH



rank 5: 2682438598672766413
 $\log M^*=10.19$ (-0.10) $z=0.762$
 PHZ $\Delta MS=+1.30$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.005$

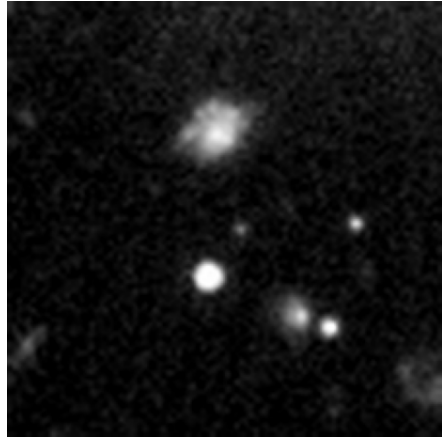


Complete candidate SFH

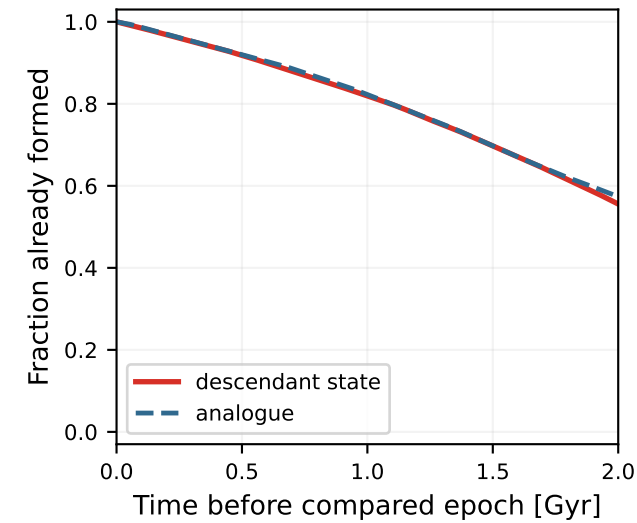
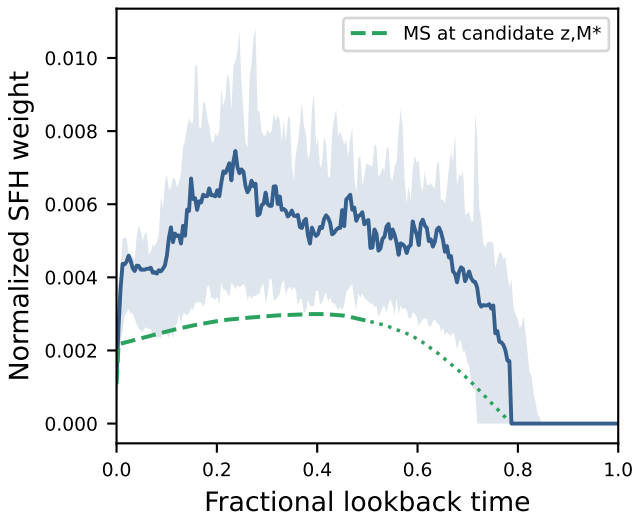


Earlier state $f=0.85$: lookback=1.69 Gyr, predicted $\log M^*=10.27$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

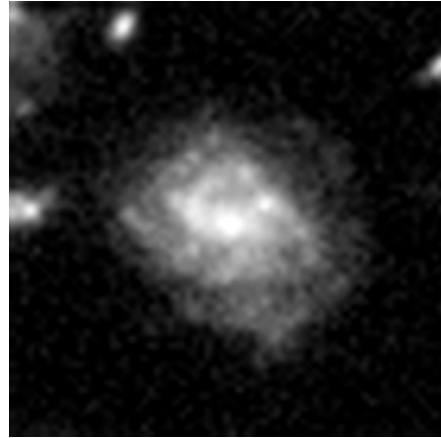
rank 1: 2710210833664871369
 $\log M^*=10.22$ (-0.05) $z=0.840$
 PHZ $\Delta MS=+0.67$; SFH-100Myr $\Delta MS=+0.21$
 $D=0.003$



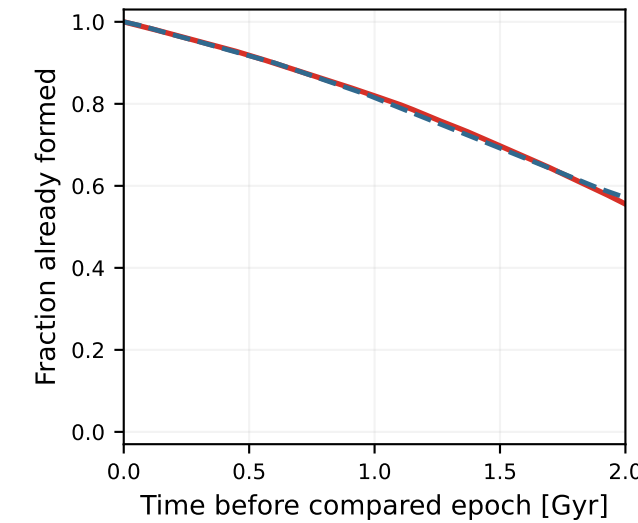
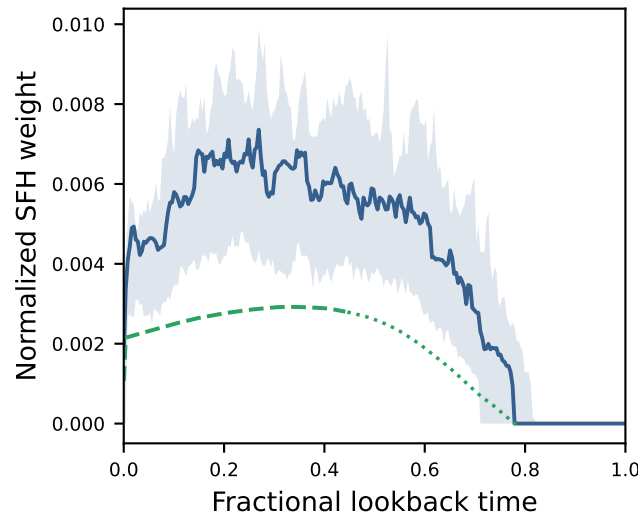
Complete candidate SFH



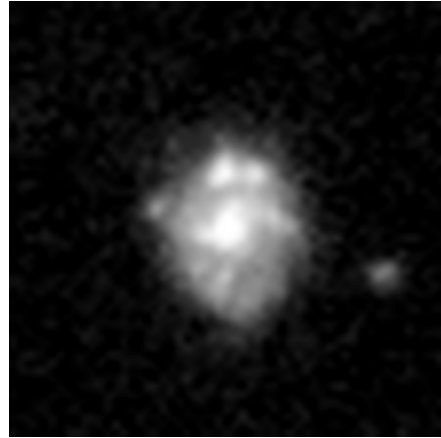
rank 2: 2666887993669930115
 $\log M^*=10.16$ (-0.11) $z=0.786$
 PHZ $\Delta MS=+1.29$; SFH-100Myr $\Delta MS=+0.26$
 $D=0.004$



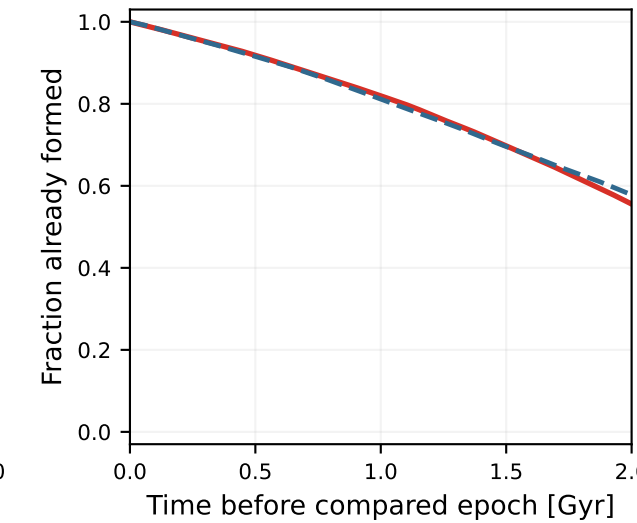
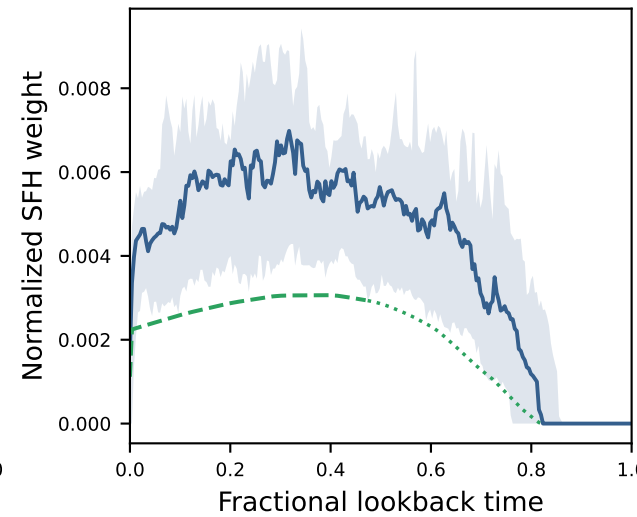
Complete candidate SFH



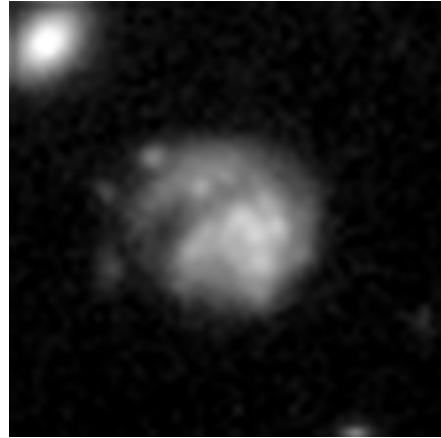
rank 3: 2762848557663621053
 $\log M^*=10.17$ (-0.10) $z=0.836$
 PHZ $\Delta MS=+1.35$; SFH-100Myr $\Delta MS=+0.23$
 $D=0.005$



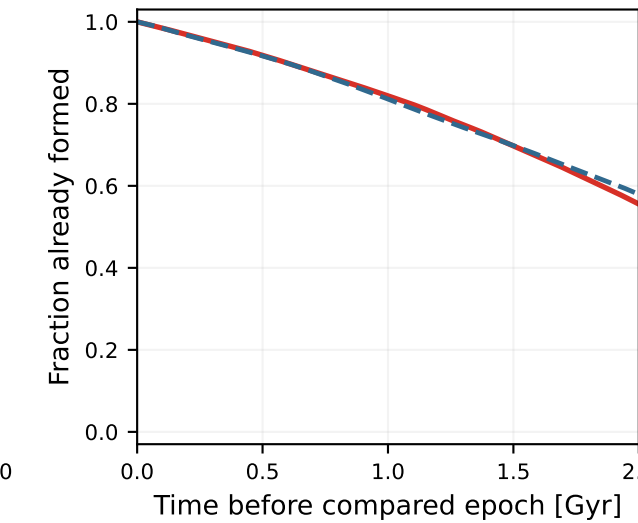
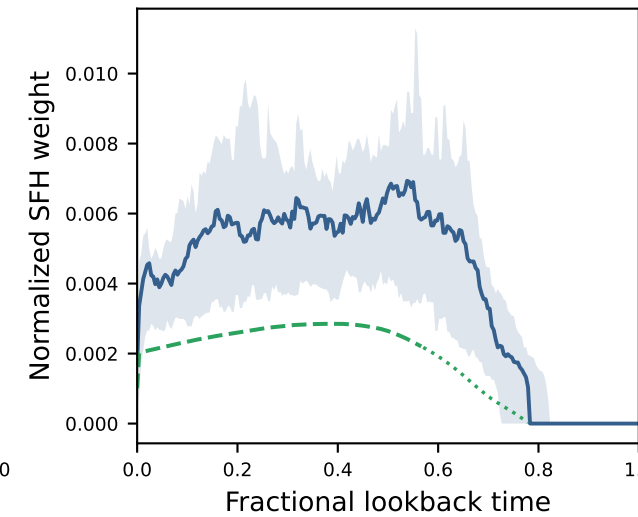
Complete candidate SFH



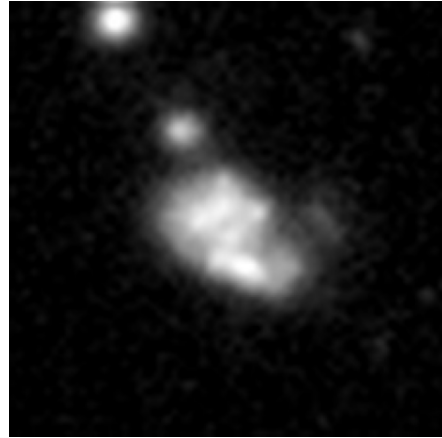
rank 4: 2754013848673523846
 $\log M^*=10.39$ (+0.12) $z=0.902$
 PHZ $\Delta MS=+1.61$; SFH-100Myr $\Delta MS=+0.27$
 $D=0.006$



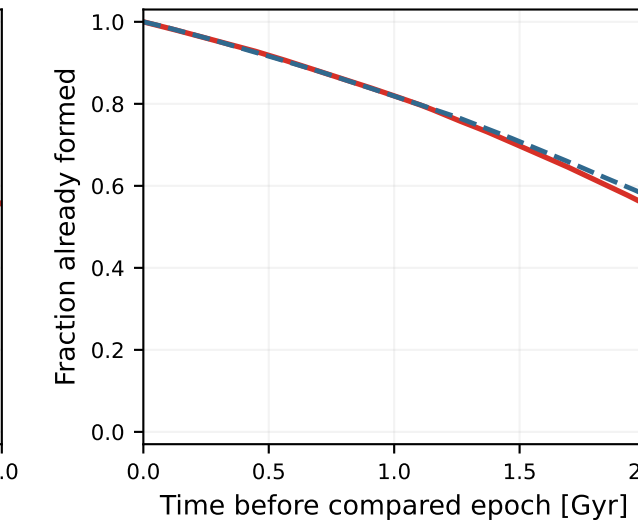
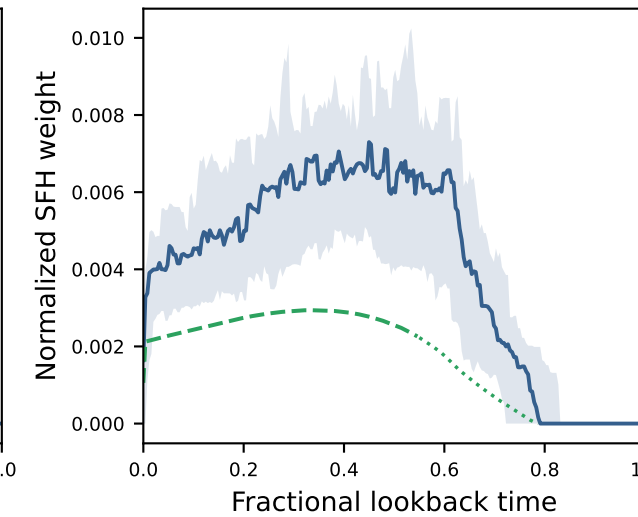
Complete candidate SFH



rank 5: 2711142278683735352
 $\log M^*=10.36$ (+0.09) $z=0.930$
 PHZ $\Delta MS=+1.56$; SFH-100Myr $\Delta MS=+0.27$
 $D=0.006$



Complete candidate SFH



Earlier state $f=0.80$: lookback=2.06 Gyr, predicted $\log M^*=10.25$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

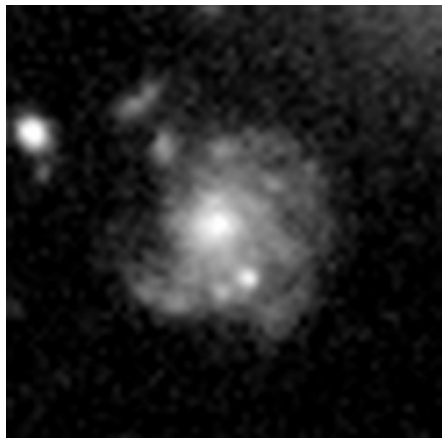
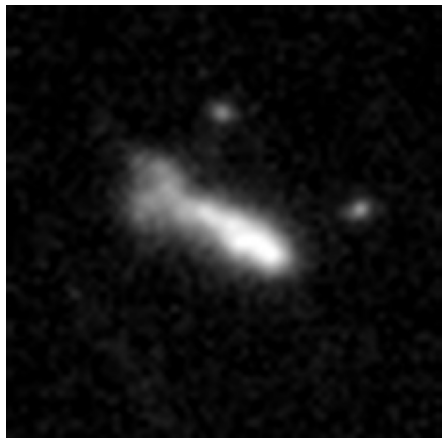
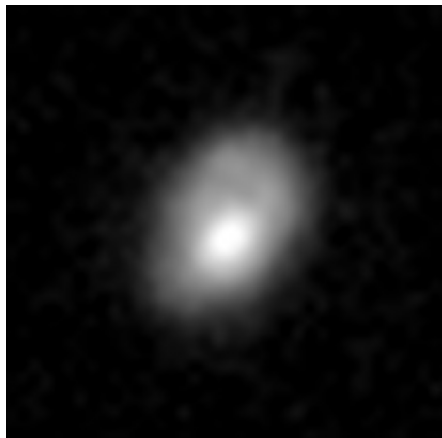
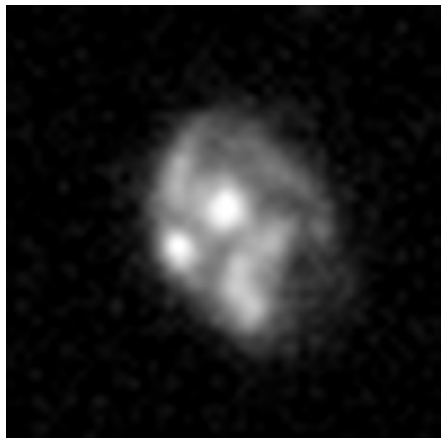
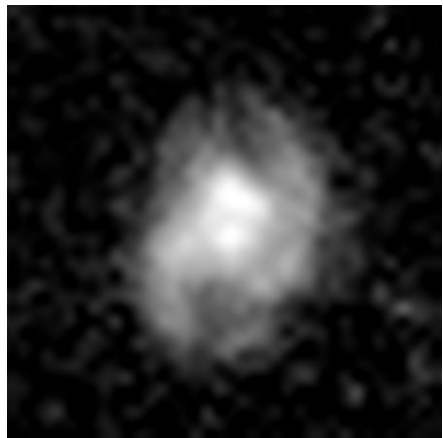
rank 1: 2678586074686170254
 $\log M^*=10.28 (+0.03)$ $z=0.978$
 PHZ $\Delta MS=+1.27$; SFH-100Myr $\Delta MS=+0.32$
 $D=0.011$

rank 2: 2641625667665171672
 $\log M^*=10.20 (-0.05)$ $z=0.988$
 PHZ $\Delta MS=+1.28$; SFH-100Myr $\Delta MS=+0.32$
 $D=0.012$

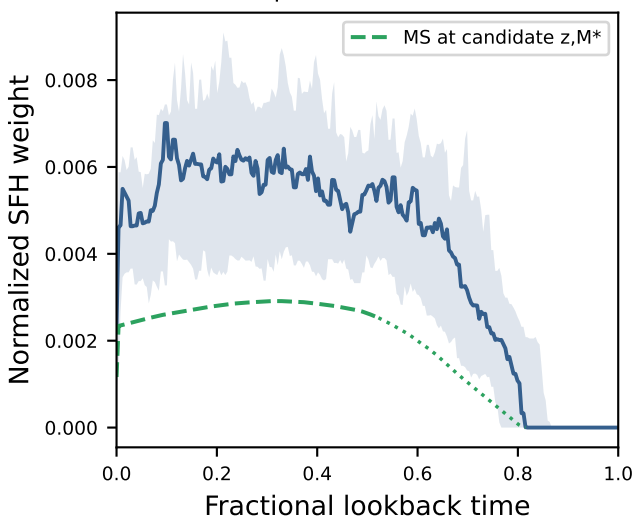
rank 3: 2660350437672187516
 $\log M^*=10.14 (-0.11)$ $z=0.728$
 PHZ $\Delta MS=+0.92$; SFH-100Myr $\Delta MS=+0.45$
 $D=0.012$

rank 4: 2638529012665704771
 $\log M^*=10.13 (-0.12)$ $z=0.930$
 PHZ $\Delta MS=+1.94$; SFH-100Myr $\Delta MS=+0.24$
 $D=0.012$

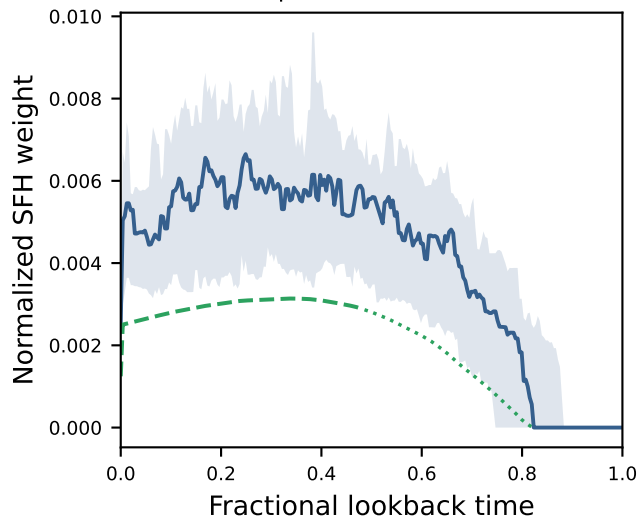
rank 5: 2655200288679618340
 $\log M^*=10.26 (+0.01)$ $z=0.902$
 PHZ $\Delta MS=+1.25$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.012$



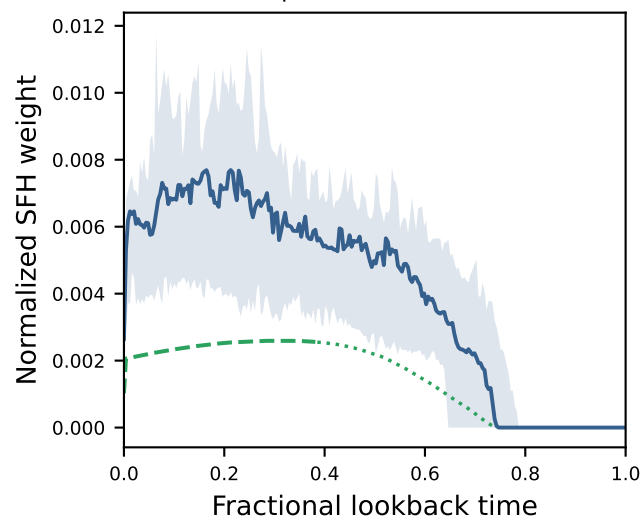
Complete candidate SFH



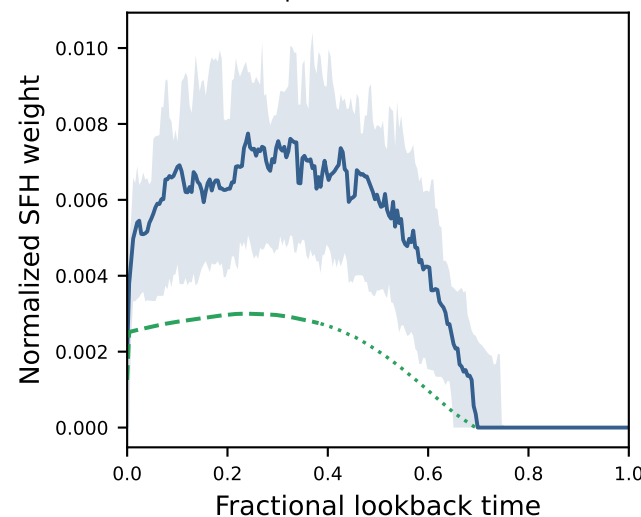
Complete candidate SFH



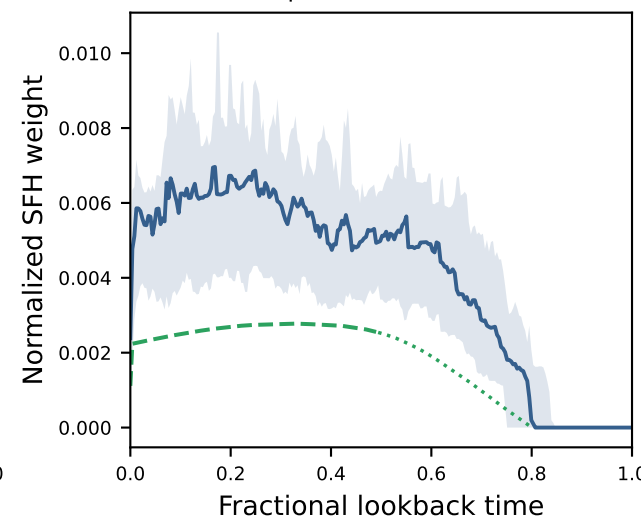
Complete candidate SFH



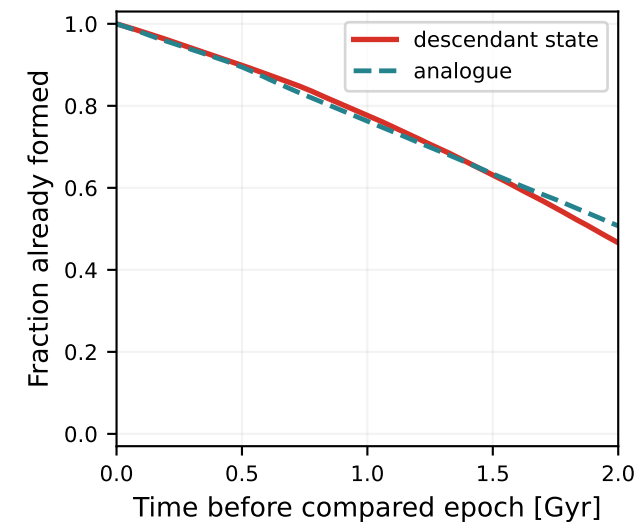
Complete candidate SFH



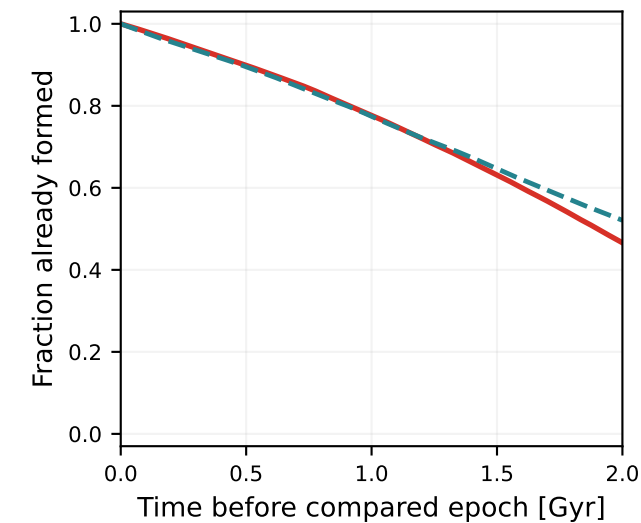
Complete candidate SFH



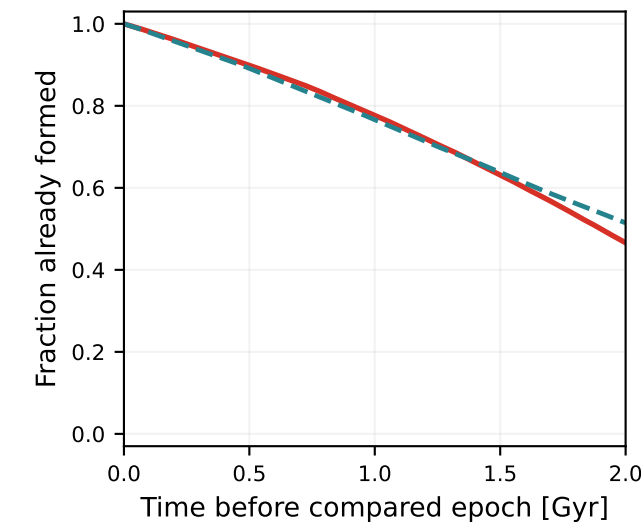
Fractional lookback time



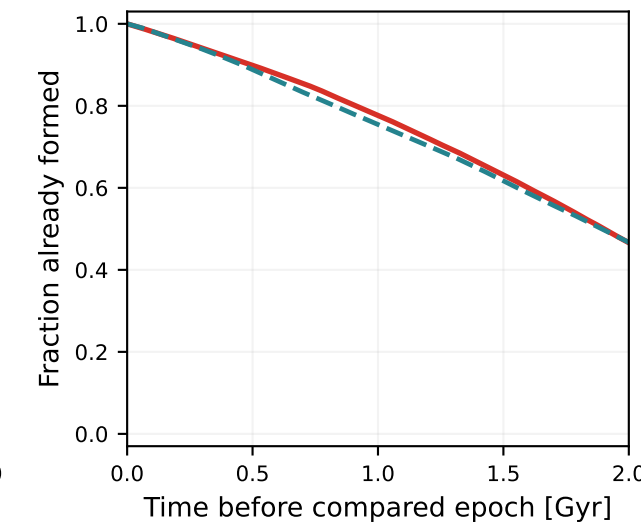
Fractional lookback time



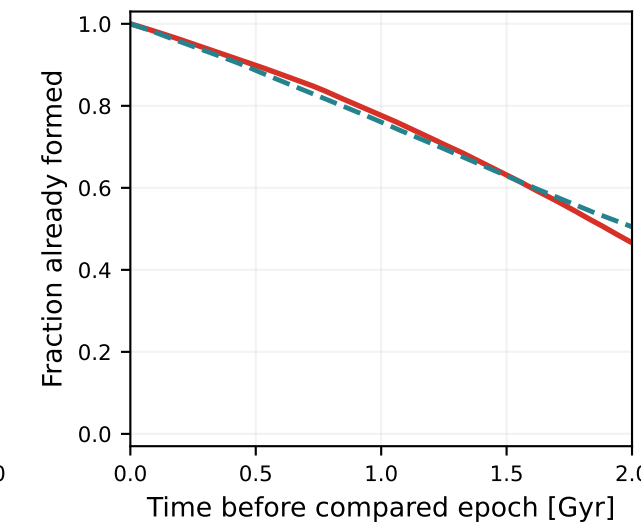
Fractional lookback time



Fractional lookback time



Fractional lookback time



Time before compared epoch [Gyr]

Time before compared epoch [Gyr]

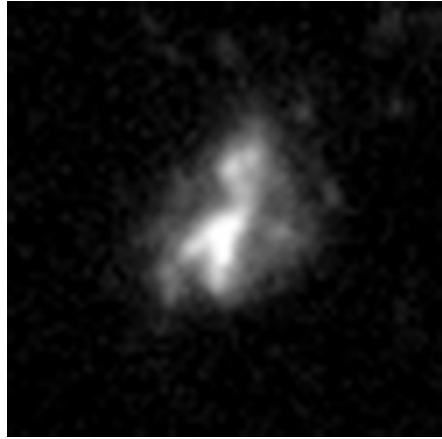
Time before compared epoch [Gyr]

Time before compared epoch [Gyr]

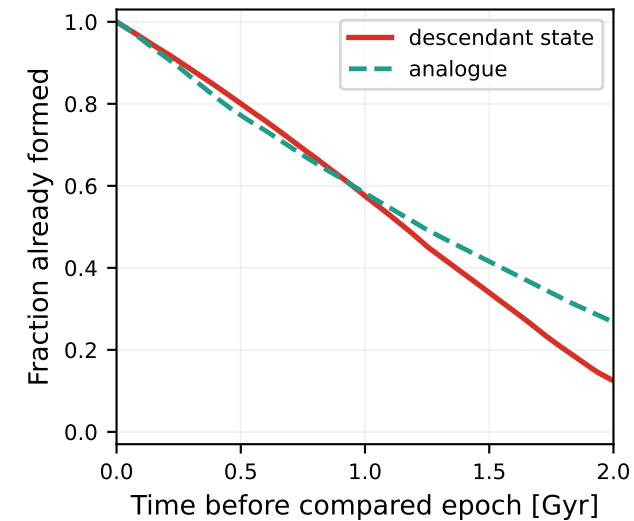
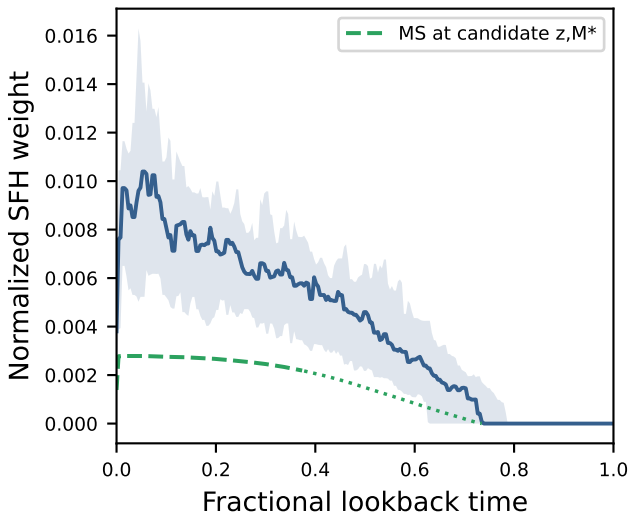
Time before compared epoch [Gyr]

Earlier state $f=0.60$: lookback=3.16 Gyr, predicted $\log M^*=10.12$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

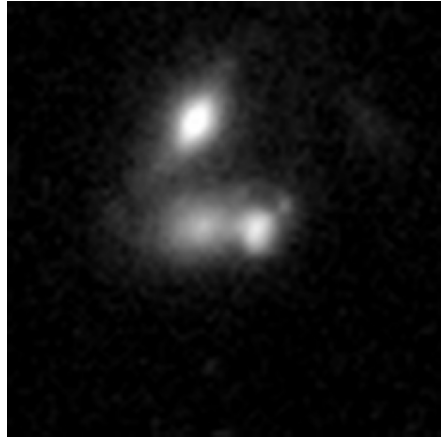
rank 1: 2717353722653902287
 $\log M^*=10.23 (+0.11)$ $z=1.198$
 PHZ $\Delta MS=+1.81$; SFH-100Myr $\Delta MS=+0.49$
 $D=0.046$



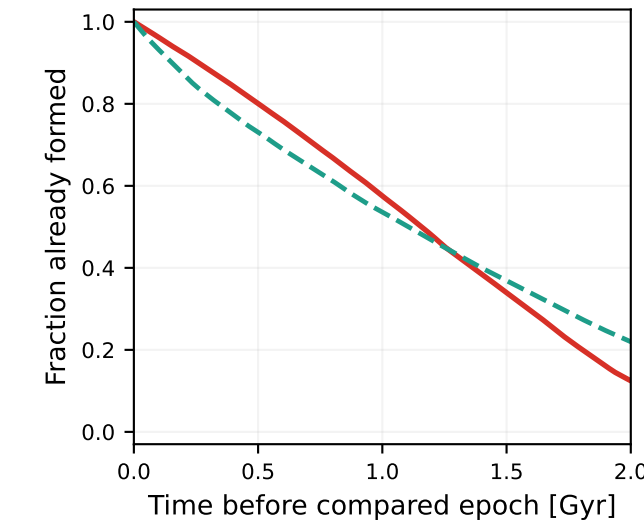
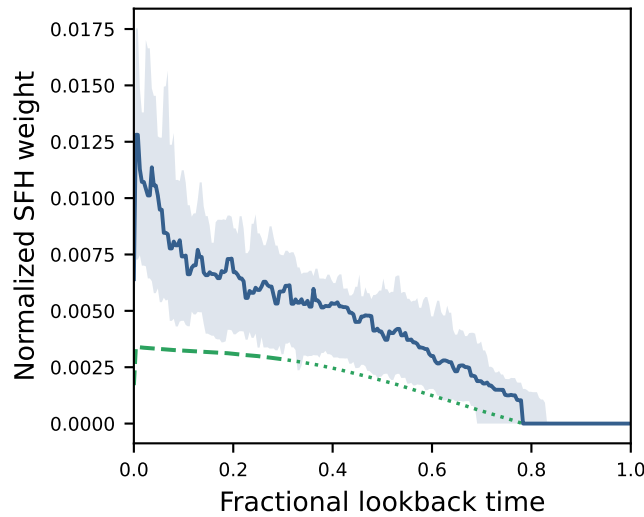
Complete candidate SFH



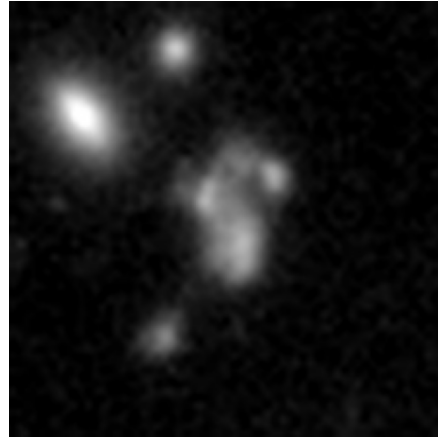
rank 2: 2658189408677769790
 $\log M^*=10.07 (-0.05)$ $z=1.464$
 PHZ $\Delta MS=+1.59$; SFH-100Myr $\Delta MS=+0.54$
 $D=0.048$



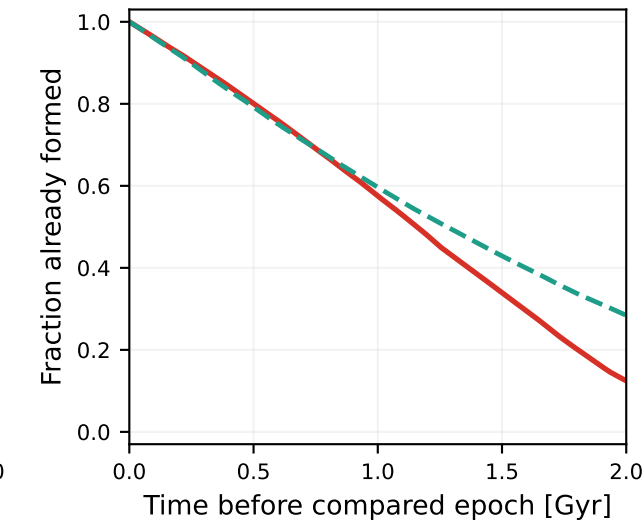
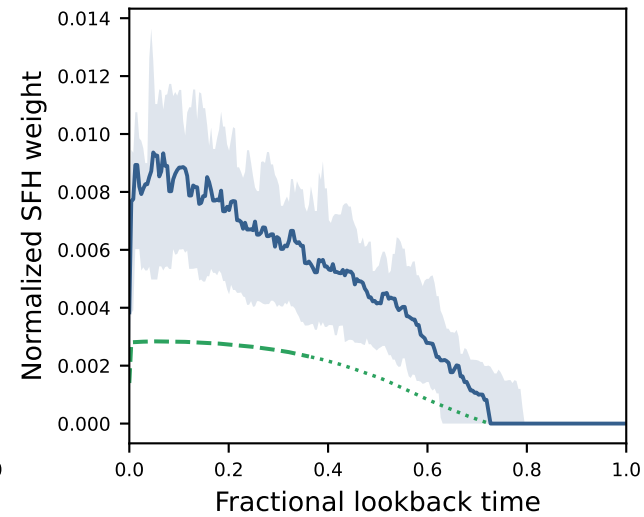
Complete candidate SFH



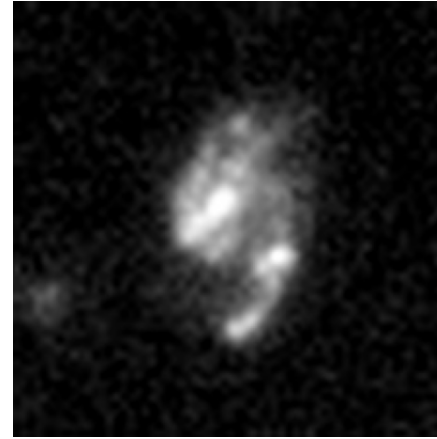
rank 3: 2737101536672032269
 $\log M^*=10.20 (+0.08)$ $z=1.174$
 PHZ $\Delta MS=+1.67$; SFH-100Myr $\Delta MS=+0.47$
 $D=0.049$



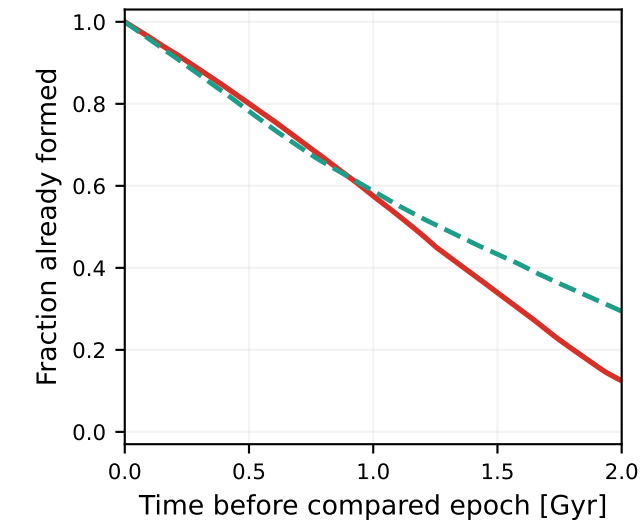
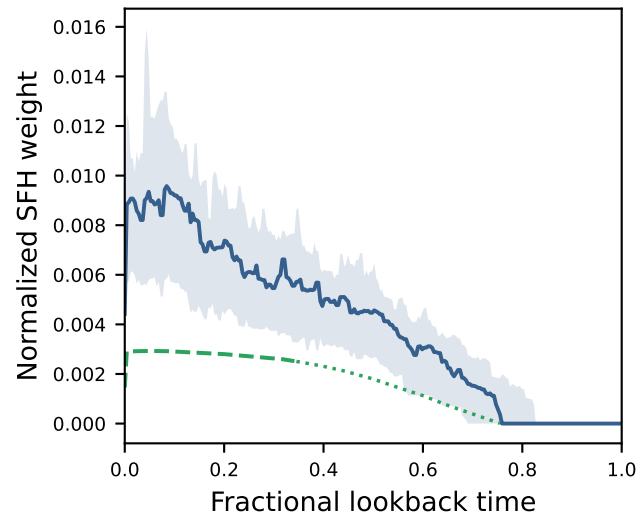
Complete candidate SFH



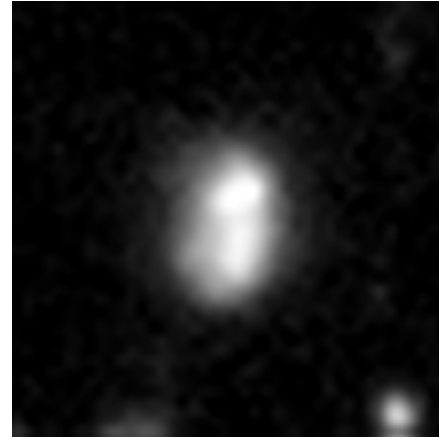
rank 4: 2691025900648883627
 $\log M^*=10.15 (+0.03)$ $z=1.182$
 PHZ $\Delta MS=+1.66$; SFH-100Myr $\Delta MS=+0.49$
 $D=0.053$



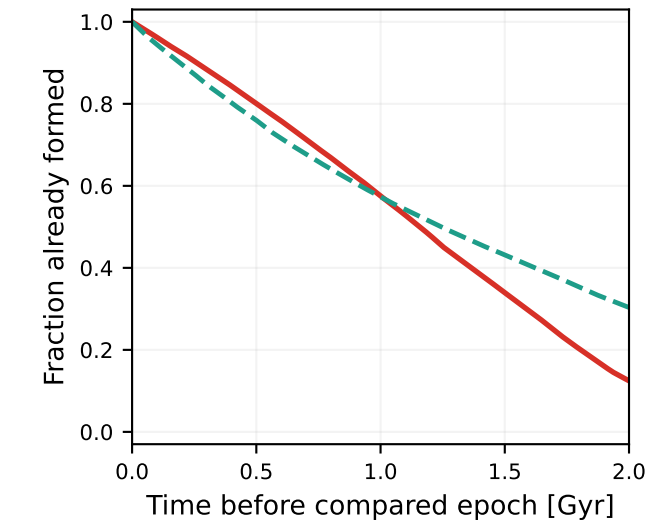
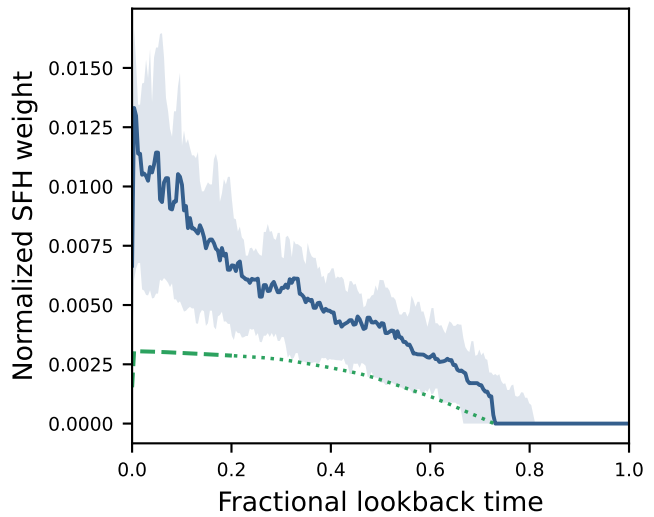
Complete candidate SFH



rank 5: 2649029123653292699
 $\log M^*=9.97 (-0.15)$ $z=1.062$
 PHZ $\Delta MS=+1.62$; SFH-100Myr $\Delta MS=+0.49$
 $D=0.061$

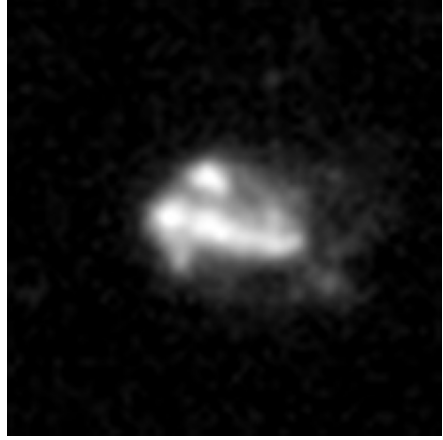


Complete candidate SFH

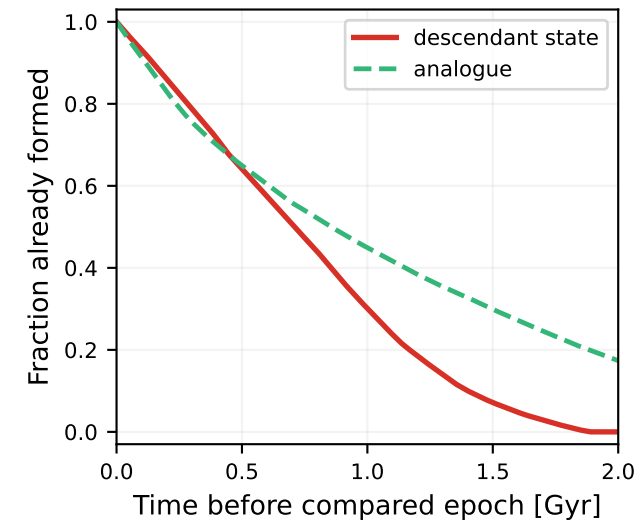
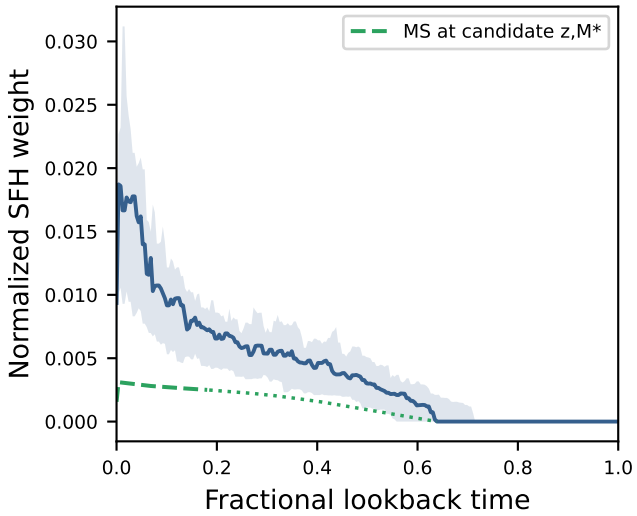


Earlier state $f=0.40$: lookback=3.96 Gyr, predicted $\log M^*=9.95$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

rank 1: 2647131718660160391
 $\log M^*=10.03 (+0.09)$ $z=1.170$
 PHZ $\Delta MS=+1.63$; SFH-100Myr $\Delta MS=+0.76$
 $D=0.126$



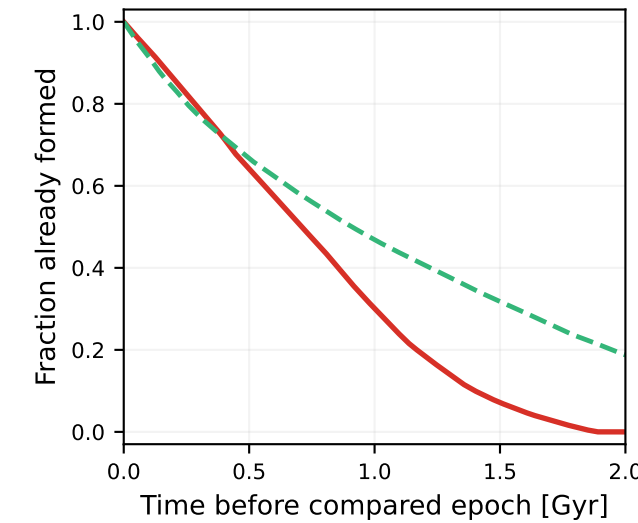
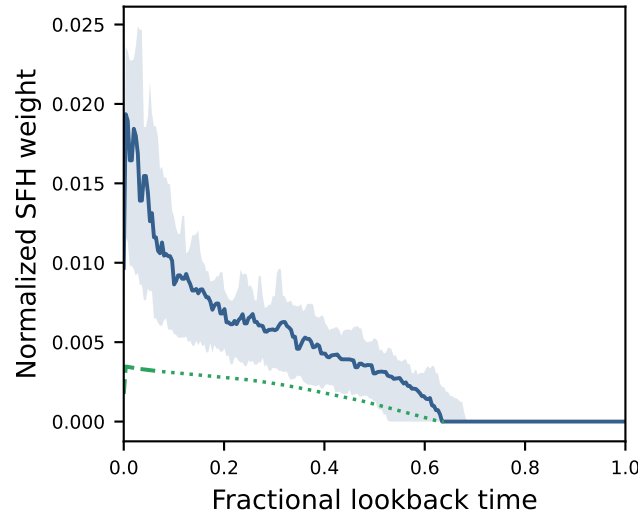
Complete candidate SFH



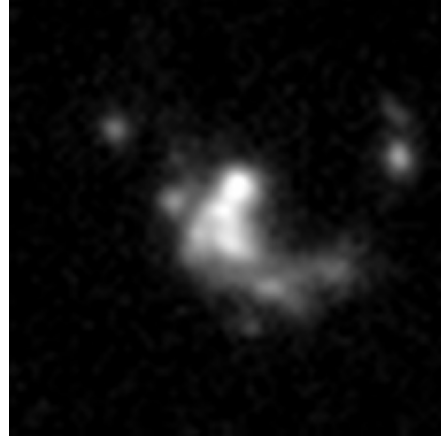
rank 2: 2766281064676170516
 $\log M^*=9.85 (-0.10)$ $z=1.138$
 PHZ $\Delta MS=+1.74$; SFH-100Myr $\Delta MS=+0.72$
 $D=0.139$



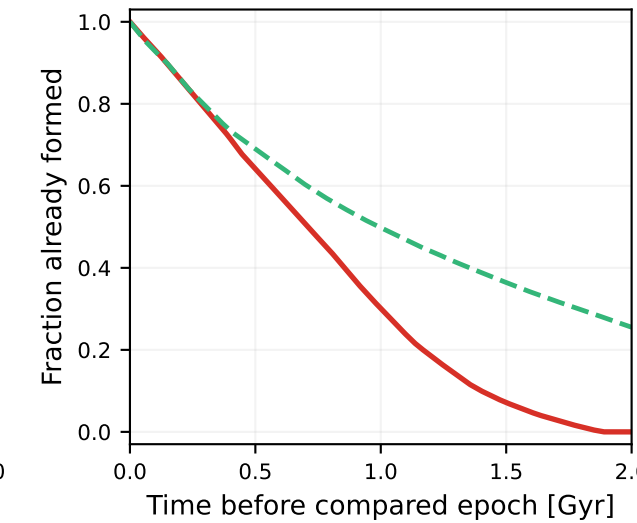
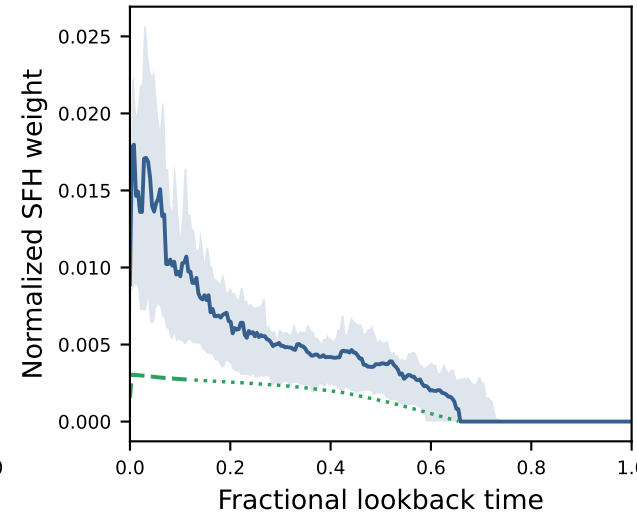
Complete candidate SFH



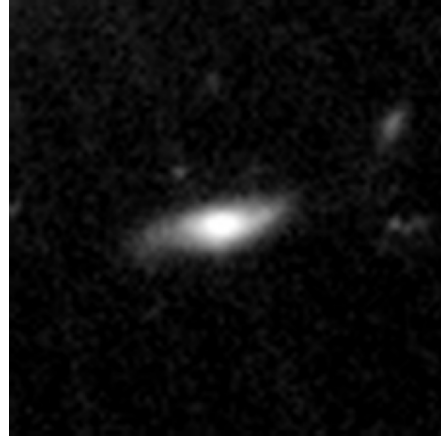
rank 3: 2650234495658951302
 $\log M^*=9.93 (-0.01)$ $z=1.010$
 PHZ $\Delta MS=+1.36$; SFH-100Myr $\Delta MS=+0.74$
 $D=0.167$



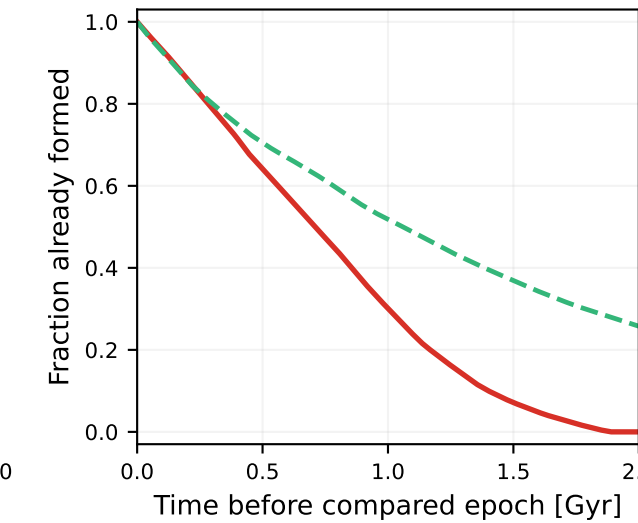
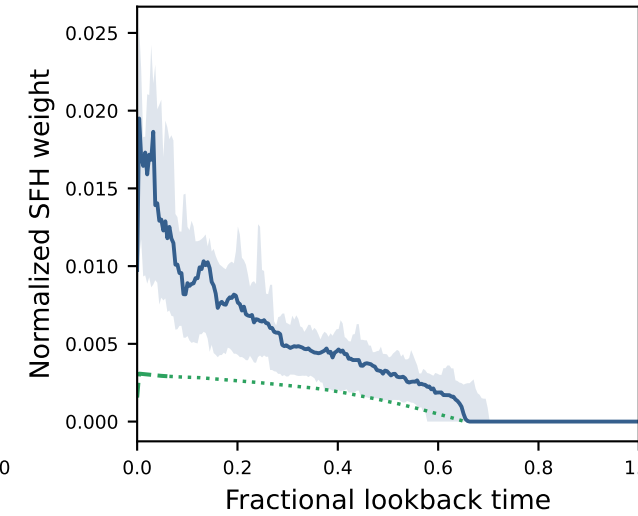
Complete candidate SFH



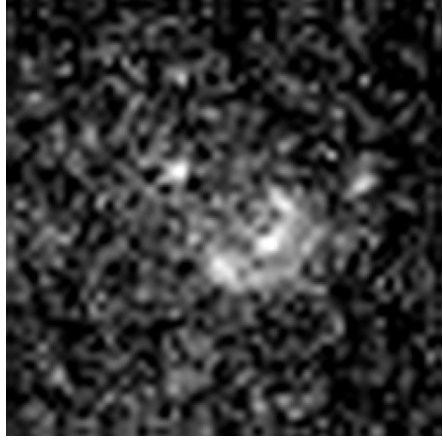
rank 4: 2708364677681344616
 $\log M^*=9.82 (-0.12)$ $z=0.920$
 PHZ $\Delta MS=+1.38$; SFH-100Myr $\Delta MS=+0.76$
 $D=0.178$



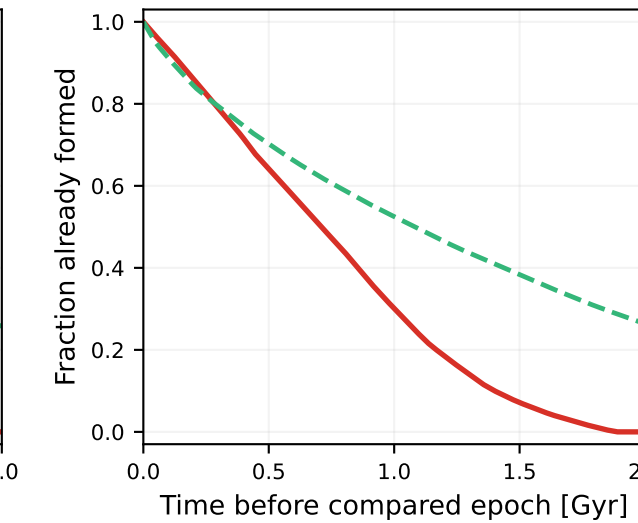
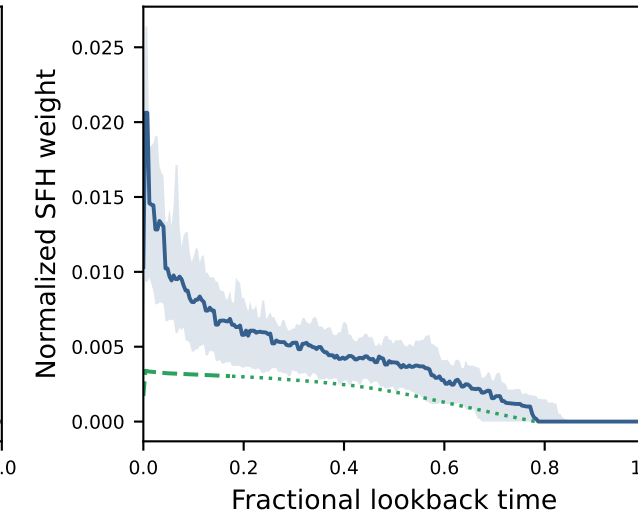
Complete candidate SFH



rank 5: 2647176063640000279
 $\log M^*=9.95 (+0.00)$ $z=1.240$
 PHZ $\Delta MS=+1.62$; SFH-100Myr $\Delta MS=+0.76$
 $D=0.184$



Complete candidate SFH



Earlier state $f=0.20$: lookback=4.67 Gyr, predicted $\log M^*=9.64$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

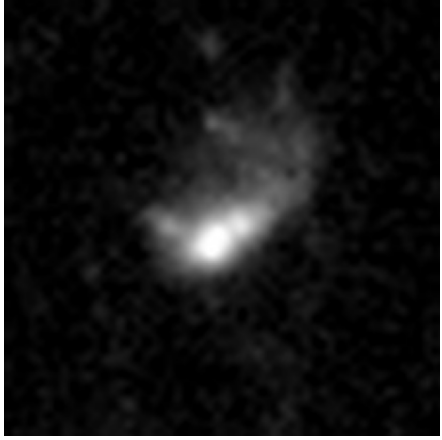
rank 1: 2645606478643808588
 $\log M^*=9.72 (+0.08)$ $z=1.116$
 PHZ $\Delta MS=+1.67$; SFH-100Myr $\Delta MS=+0.83$
 $D=0.259$

rank 2: 2744318378668883710
 $\log M^*=9.65 (+0.00)$ $z=0.770$
 PHZ $\Delta MS=+1.31$; SFH-100Myr $\Delta MS=+0.98$
 $D=0.277$

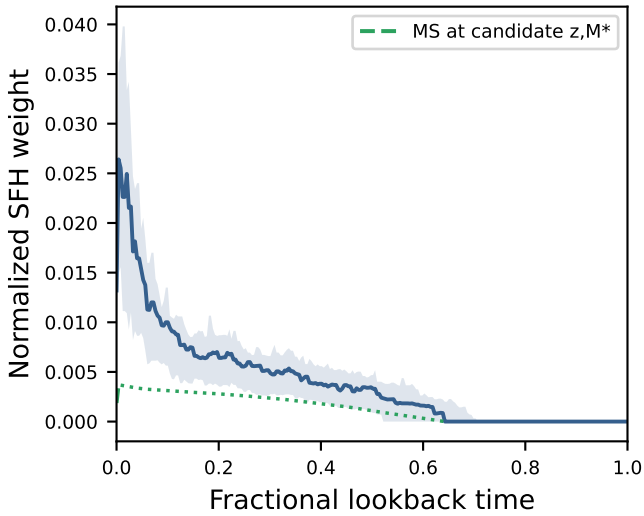
rank 3: 2733550546657853038
 $\log M^*=9.76 (+0.11)$ $z=0.760$
 PHZ $\Delta MS=+1.25$; SFH-100Myr $\Delta MS=+0.97$
 $D=0.305$

rank 4: 2695901997638798783
 $\log M^*=9.55 (-0.10)$ $z=1.234$
 PHZ $\Delta MS=+1.48$; SFH-100Myr $\Delta MS=+0.50$
 $D=0.308$

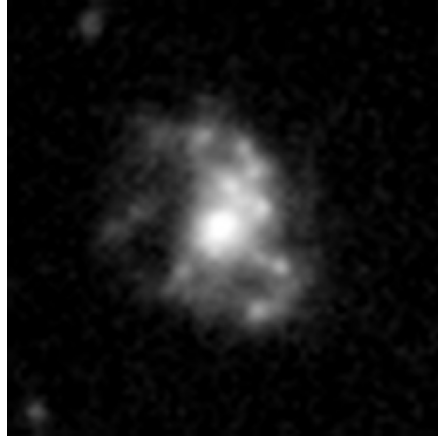
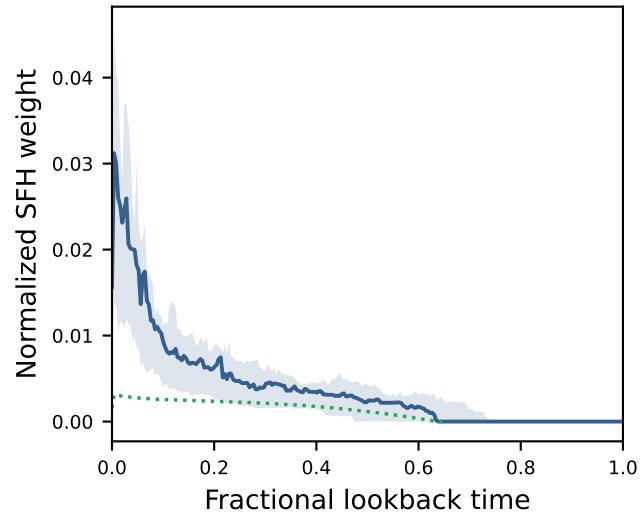
rank 5: 2753054265674679912
 $\log M^*=9.50 (-0.14)$ $z=0.560$
 PHZ $\Delta MS=+1.19$; SFH-100Myr $\Delta MS=+1.0$
 $D=0.335$



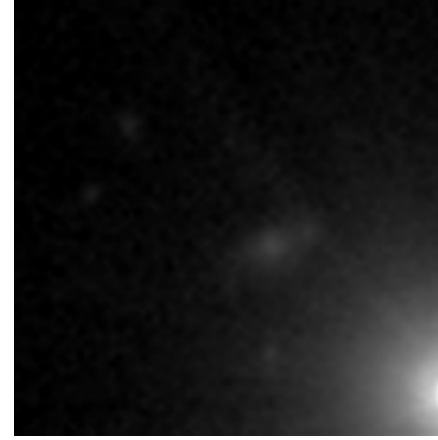
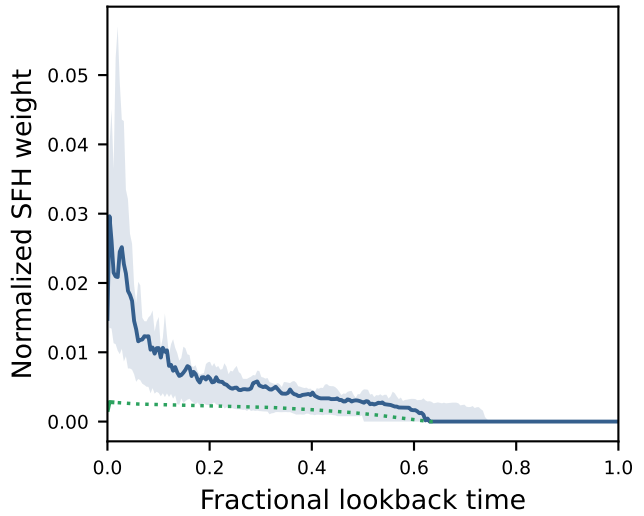
Complete candidate SFH



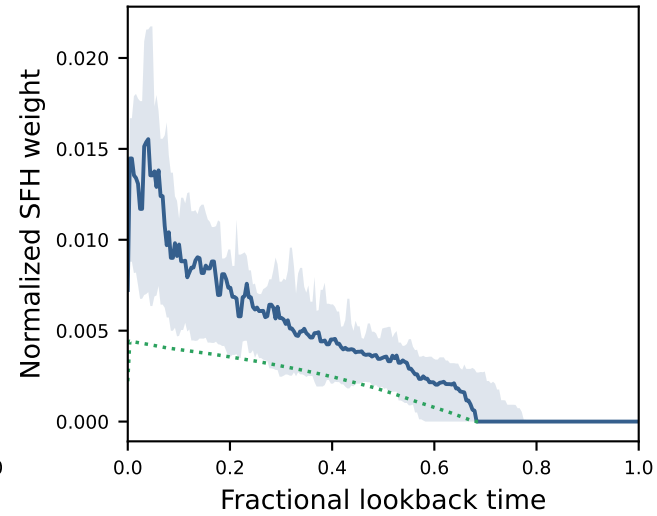
Complete candidate SFH



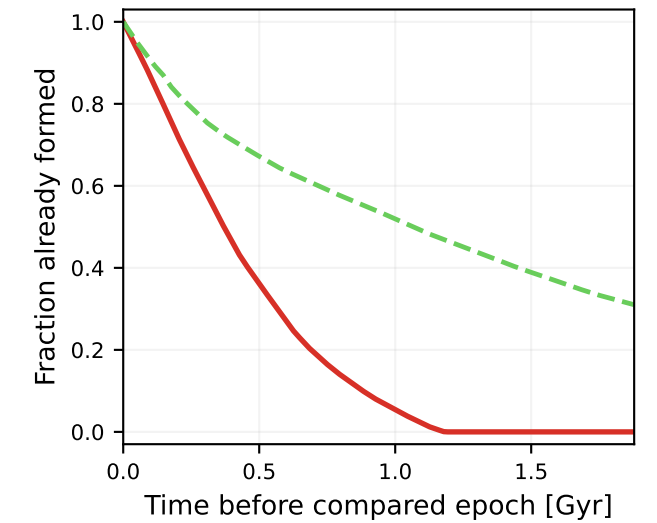
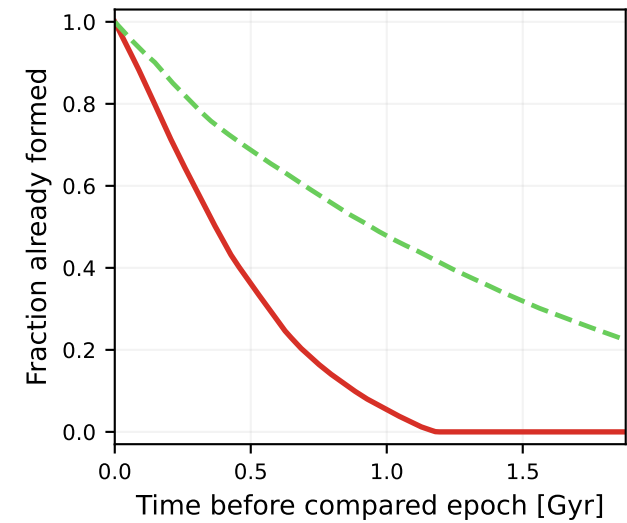
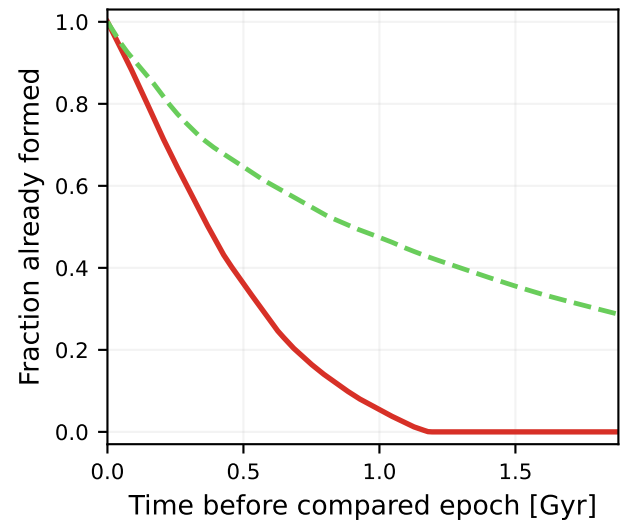
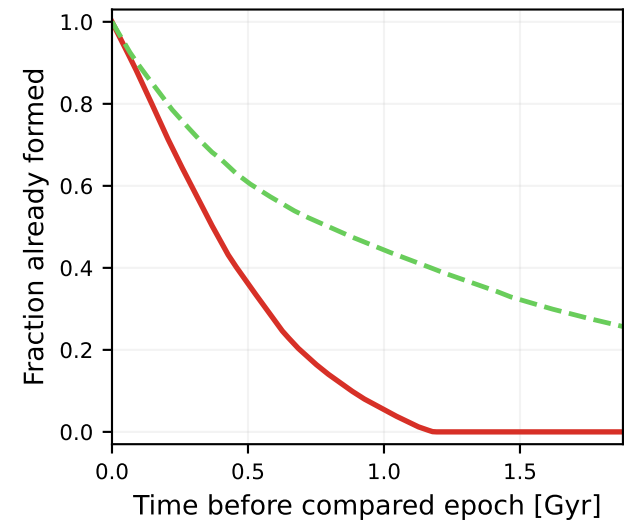
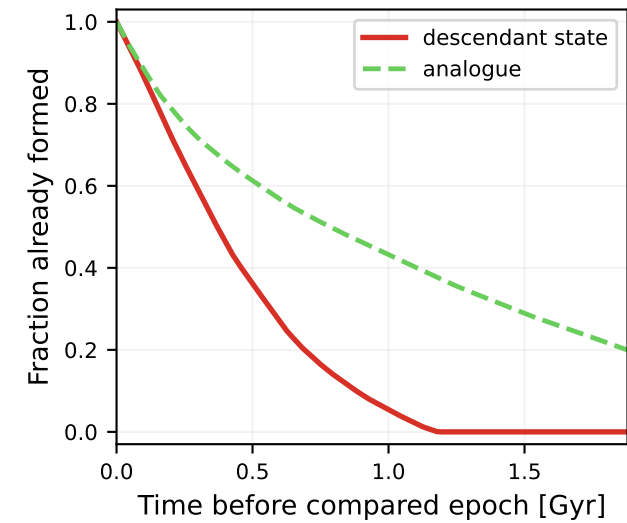
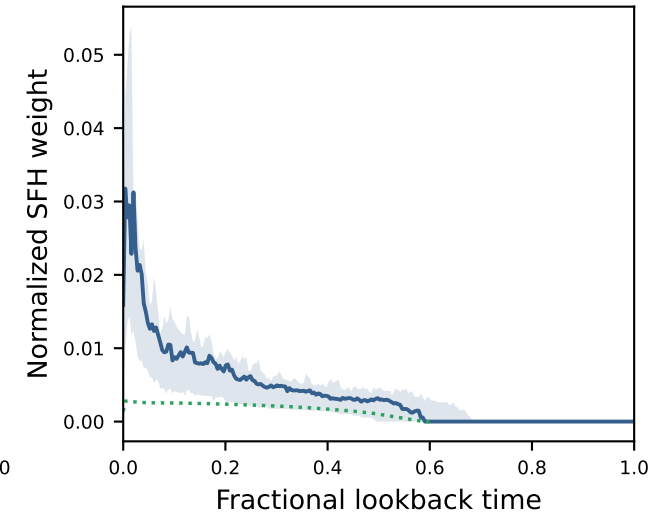
Complete candidate SFH



Complete candidate SFH



Complete candidate SFH

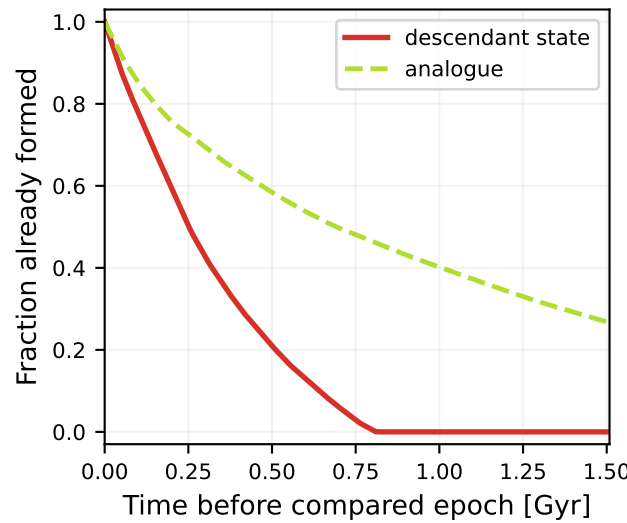
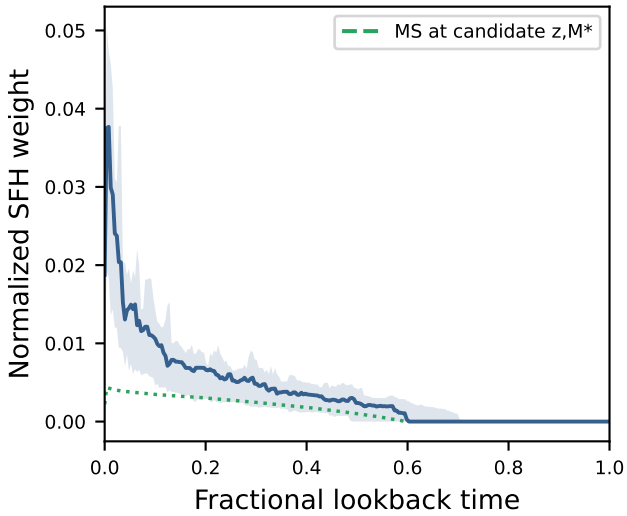


Earlier state $f=0.10$: lookback=5.04 Gyr, predicted $\log M^*=9.34$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

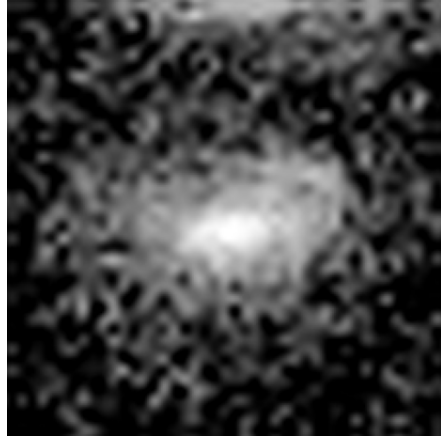
rank 1: 2648545772681764910
 $\log M^*=9.39 (+0.05)$ $z=0.994$
 PHZ $\Delta MS=+1.79$; SFH-100Myr $\Delta MS=+0.90$
 $D=0.320$



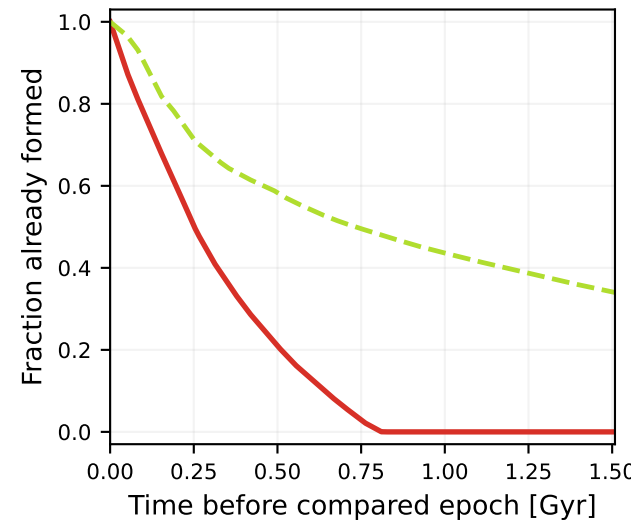
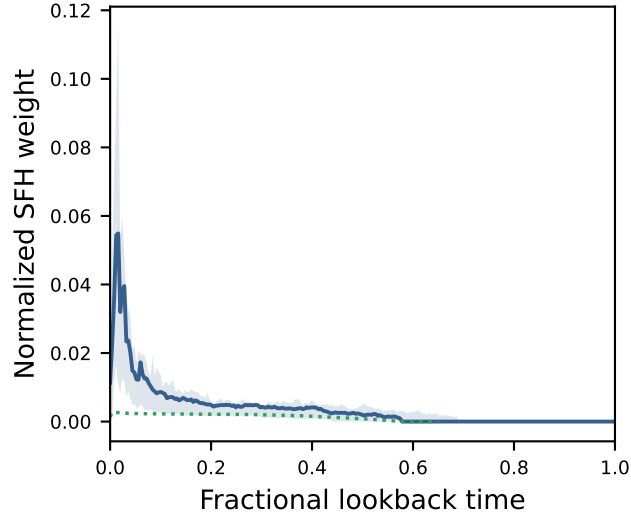
Complete candidate SFH



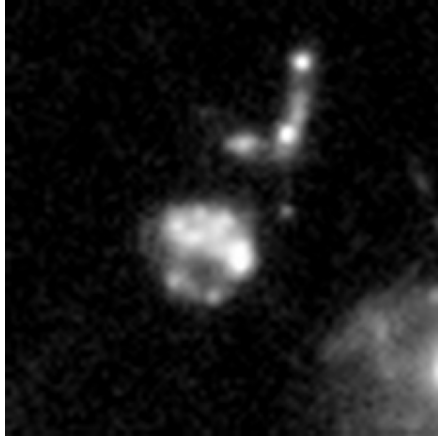
rank 2: 2647975483685285807
 $\log M^*=9.44 (+0.09)$ $z=0.510$
 PHZ $\Delta MS=+1.34$; SFH-100Myr $\Delta MS=+1.07$
 $D=0.346$



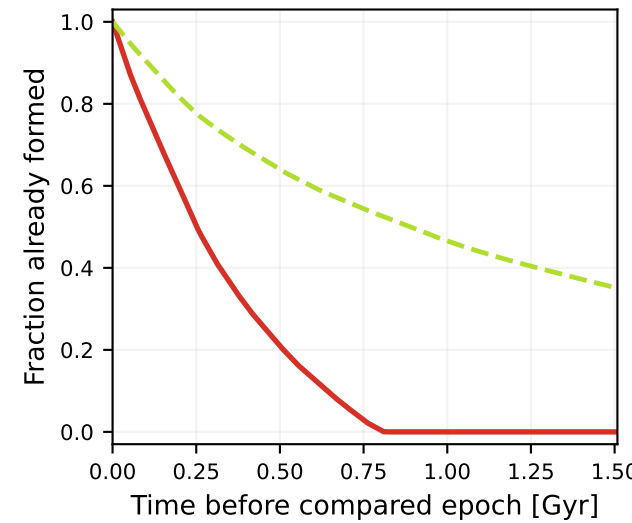
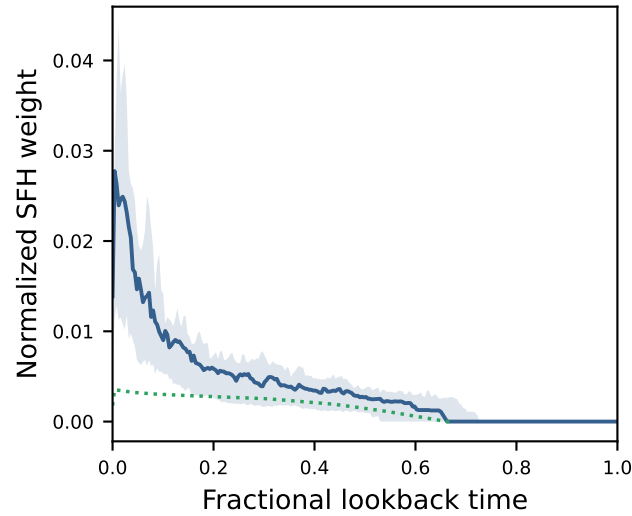
Complete candidate SFH



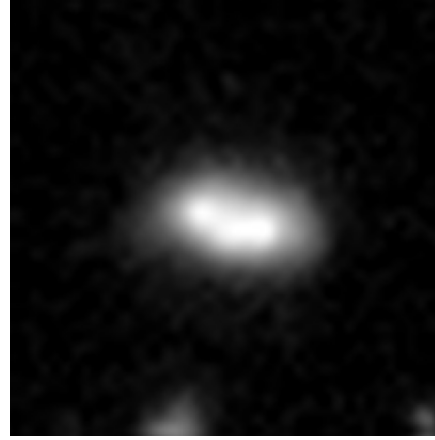
rank 3: 2740268308665100250
 $\log M^*=9.47 (+0.12)$ $z=0.760$
 PHZ $\Delta MS=+1.77$; SFH-100Myr $\Delta MS=+0.87$
 $D=0.382$



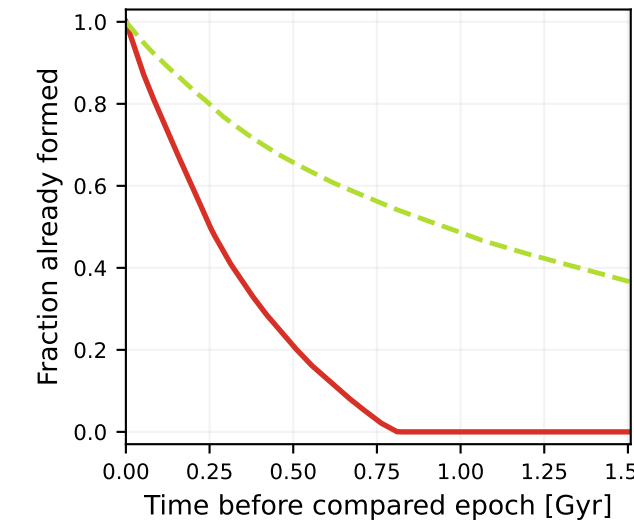
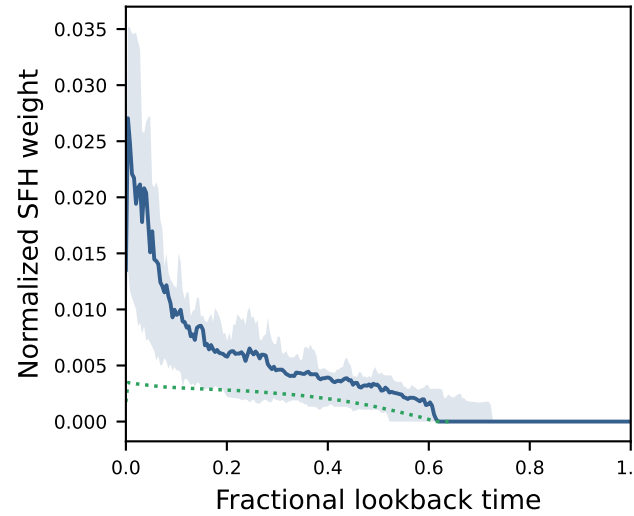
Complete candidate SFH



rank 4: 2729264667640595011
 $\log M^*=9.49 (+0.15)$ $z=0.760$
 PHZ $\Delta MS=+1.78$; SFH-100Myr $\Delta MS=+0.86$
 $D=0.398$



Complete candidate SFH



rank 5: 2758021642653704394
 $\log M^*=9.38 (+0.04)$ $z=0.950$
 PHZ $\Delta MS=+1.81$; SFH-100Myr $\Delta MS=+0.7$
 $D=0.421$



Complete candidate SFH

